

# Near-miss constructions, interaction surfaces, and prime-support obstructions for the $3 \times 3$ magic square of squares\*

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## Abstract

Whether a  $3 \times 3$  magic square of nine distinct positive perfect squares exists is a well-known open problem. This paper studies the nearest structured objects along three axes. Call a  $3 \times 3$  integer array an  $(8, 7)$  *array* if its nine entries are positive and pairwise distinct, exactly eight of them are perfect squares, and exactly seven of its eight ordinary lines (rows, columns, long diagonals) share one common sum; every use of “ $(8, 7)$  near miss” below refers to this metric. We also classify specified  $(9, 7)$  arrays, with all nine entries square. First, we give a self-contained normal form — every seven-line array is three arithmetic progressions with one common difference — together with the integral Smith form of its line-sum map, and construct infinite  $(8, 7)$  families by three mechanisms: an antidiagonal retraction of the catalogued Bremner–Wesolowski six-square carriers with a complete least-shift law, a bridge completion in Fituvalu’s type-6.VI frame, and an elliptic-orbit lift on  $E_{840}$  giving a positive-density sequence of pairwise inequivalent entry-primitive examples. Second, closing the eighth line is shown to be a rational-point problem on one arithmetic surface: a single linear condition on  $E_{840}^3$ , whose saturated model is an irreducible surface carrying an  $S_5$ -monodromy cone map; one fibre of the interaction is closed completely, fixed-squareclass finiteness follows from an imported theorem of Caro and García-Fritz, and the route through finite Kummer classes alone is provably insufficient. Third, an  $S$ -unit law for the common difference  $d$  of the three progressions bounds the number of configurations for every fixed prime support and excludes whole support classes. A consequence of Aebi’s classification gives at most one positive square progression for any difference with at most three prime factors. Using the complete  $S$ -unit totals of von Känel–Matschke, an exact catalogue excludes fully magic squares whose transversal difference has only primes at most 19, and classifies the 384 associated primitive  $(9, 7)$  classes. Arbitrary four-prime supports remain open. The companion repository supplies exact Python replays for the elementary and symbolic computations, and hash-pinned inputs and normalized output records for the external computer-algebra computations.

## Setting and objects

The following objects are kept distinct. An  $(8, 7)$  *array* is as in the abstract; a  $(9, 7)$  *array* has nine distinct positive square entries and exactly seven equal ordinary line sums. A *six-square carrier* is a fully magic  $3 \times 3$  array of six squares and three non-squares from the catalogued classical families. A *Parker configuration* is a triple of pairwise-distinct three-term progressions of perfect squares with one *common* difference  $d$ , the structure forced on the three transversals of a carrier by the completion problem. In the finiteness and support-law statements, the configuration additionally satisfies the full-magic interaction  $s_1^2 + s_2^2 = 2s_0^2$ , with the positive, distinct-entry open

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\*No novelty or priority is asserted anywhere in this document; every classical ingredient is attributed in Section 1 and in the companion repository’s source lock, and each no-hit prior-art search is reported as a dated negative search, never as a certificate.

locus imposed. A general  $(9, 7)$  array does not satisfy this interaction. The support statements concern  $\text{supp}(2d)$  — the prime support of the common difference — and never the prime factors of the entries of a hypothetical completed square, which is the object studied by Rabern, Woll, Labruna and Weisenberg. The label recalls Matt Parker’s widely known Parker Square [Par]; it does not attribute the underlying problem to Parker, and Parker’s particular near miss is not a configuration studied in this paper.

## Main results

**Theorem 1** (Seven-line normal form). *Up to the dihedral symmetry of the grid, every  $3 \times 3$  integer array in which exactly seven of the eight ordinary lines share a common sum has the antidiagonal as its failed line and is exactly*

$$\begin{pmatrix} A & P+D & E-D \\ P-D & E & A+D \\ E+D & A-D & P \end{pmatrix},$$

*three arithmetic progressions of one common difference  $D$ ; the seven lines sum to  $A+E+P$  and the failed diagonal to  $3E$ . Consequently an  $(8, 7)$  array consists of two complete square progressions and one progression containing exactly two squares. The integral line-sum map  $\Lambda : \mathbb{Z}^9 \rightarrow \mathbb{Z}^8$  has Smith form  $\text{diag}(1, 1, 1, 1, 1, 1, 3, 0)$ ; hence its cokernel is  $\mathbb{Z} \oplus \mathbb{Z}/3\mathbb{Z}$  and characteristic 3 is the unique prime at which its rank drops.*

**Theorem 2** (Infinite  $(8, 7)$  families). (i) *Subtracting  $6z_n^2$  from the antidiagonal of the  $n$ -th catalogued Bremner–Wesolowski carrier yields, for every  $n \geq 1$ , an  $(8, 7)$  array with explicit square entries and defect  $12z_n^2$ ; every odd-indexed member is root-primitive.* (ii) *For fixed  $n$ , all positive shifts  $0 < t < C$  closing at least one antidiagonal pair to squares are classified by same-parity factor pairs of  $\Delta$  and  $2\Delta$ , and the least such shift is 9 for  $n = 1$  and  $6z_n^2$  for every  $n \geq 2$ ; it is not unique.* (iii) *In the type-6.VI frame the fixed bridge  $(23^2, 37^2, 47^2)$  of difference 840 admits exactly the tautological completion  $(23, 37, 47)$  and one further completion  $(79, 65, 89)$ ; coupled with Robertson’s progression  $(1^2, 29^2, 41^2)$  it produces an explicit  $(8, 7)$  array.* (iv) *The non-torsion orbit of  $P = (841, -1189)$  on  $E_{840} : Y^2 = X^3 - 840^2 X$  re-enters the admissible window infinitely often and yields infinitely many pairwise square-scale/ $D_4$ -inequivalent, entry-primitive  $(8, 7)$  arrays from Bremner’s Configuration VI. More precisely, the admissible indices have natural density*

$$\delta = 1 - I_{(840/4225)^2} \left( \frac{1}{4}, \frac{1}{2} \right) = 0.6585271498271213 \dots$$

**Theorem 3** (The interaction surface). (i) *After independent common subtractions  $\tau_0, \tau_1, \tau_2$  on the three difference-840 transversals of the fixed carrier, all rows and columns remain equal and the two diagonals agree if and only if  $\tau_1 + \tau_2 = 2\tau_0$ ; equivalently  $x_1 + x_2 = 2x_0$  with each  $x_i$  in  $x(2E_{840}(\mathbb{Q}))$  — one equation on  $E_{840}^3$ .* (ii) *Saturating the associated polynomial system by every zero, denominator and collision divisor leaves one irreducible surface of dimension two, which maps onto the rational cone  $2A^2 = B^2 + C^2$  with generic degree five in squared coordinates (degree forty on the full signed cover) and generic monodromy  $S_5$ .*

**Theorem 4** (Finiteness and one closed fibre). (i) **Derived from an imported theorem** (Caro–García-Fritz, Thms. 1.2–1.3): *for fixed  $d$ , the points of  $(2E_d(\mathbb{Q}))^3$  satisfying the interaction, after removing points at infinity, nonpositive and repeated  $x$ -coordinates, form a finite set, with the imported bound of shape  $D^{3r+1}$ ,  $r = \text{rank } E_d(\mathbb{Q})$ ; hence for every fixed squareclass of  $d$  only finitely*

many positive distinct Parker configurations exist modulo rational-square scaling. The bound is non-effective. (ii) **Computer-assisted** (the decisive rank and torsion enter as a hash-verified frozen transcript of the official Magma Calculator): the fibre  $x_0 = 841$  is closed completely — its necessary genus-two curve has a complementary bielliptic quotient of rank zero and full two-torsion, whose four rational points pull back to  $t \in \{0, \pm 1, \pm 841, \pm 1681\}$ , and positivity leaves only  $t = 0$ , so the fibre contains no positive nonconstant three-transversal completion.

**Theorem 5** (The finite-class stop). *Half classes  $\lambda_n(P) = [Q] \in E_d(\mathbb{Q})/(2^n E_d(\mathbb{Q}) + E_d(\mathbb{Q})[2])$  are well defined for  $P \in 2E_d(\mathbb{Q})$ . On  $E_{840}$ , neither  $P \mapsto x(P)$  nor the interaction of Theorem 3 factors through the finite class triple for any  $n \geq 1$ : the explicit pair  $Q, Q + 2^n R$  has equal classes and unequal doubled  $x$ -coordinates. Hence no obstruction retaining only the three individual finite Kummer/2-Selmer classes can characterize the interaction. The certified true witness is the repeated triple  $(Q, Q, Q)$ ; a stronger statement whose hypotheses first impose the positive pairwise-distinct open locus is not excluded.*

**Theorem 6** (The support law). *For a positive pairwise-distinct Parker configuration put  $u_i = t_i - r_i$ ,  $v_i = t_i + r_i$ ,  $z_i = u_i/v_i$ . Then  $u_i v_i = 2d$ ,  $0 < z_i < 1$ , and the interaction is exactly the five-term unit equation*

$$z_1 + z_1^{-1} + z_2 + z_2^{-1} = 2(z_0 + z_0^{-1}),$$

*nondegenerate on the Parker open. For every finite prime set  $\mathcal{S} \ni 2$ , the number of root-primitive configurations with  $\text{supp}(2d) \subseteq \mathcal{S}$ , modulo common rational-square scaling, is at most*

$$\exp(30^{15}(3|\mathcal{S}| + 1)).$$

*This counting theorem by itself produces no enumeration; the separate fixed-support route is given in Section 8. At every odd prime the largest of  $|v_p(z_i)|$  occurs at least twice; the admissible endpoint pairs obey the explicit  $(-1/p), (2/p)$  table; and  $\text{supp}(2d) \subseteq \{2, 3\}$  is impossible for a primitive configuration.*

**Theorem 7** (Complete three-prime exclusion). **Consequence of Aebi's classification.** *For any positive integer  $d$  with at most three distinct prime divisors, there is at most one positive integer triple  $r < s < t$  such that  $s^2 - r^2 = t^2 - s^2 = d$ . No primitivity assumption is required. In particular, neither a  $(9, 7)$  array nor a fully magic square of nine distinct positive integer squares can have such a transversal difference. The primitive Pythagorean-triangle classification is Aebi's; this statement is its squareclass consequence, without a priority claim.*

**Theorem 8** (Smooth differences and a complete  $(9, 7)$  atlas). **Computer-assisted, using imported complete  $S$ -unit totals.** *Let  $S_B$  be the primes at most  $B$ . (i) There is no fully magic  $3 \times 3$  square of nine distinct positive integer squares whose transversal difference  $d$  satisfies  $\text{supp}(d) \subseteq S_{19}$ . (ii) The  $(9, 7)$  arrays with  $\text{supp}(|D|) \subseteq S_B$ , modulo  $D_4$  and common rational-square scaling, have exactly 3,45,384 classes for  $B = 7, 13, 19$ , respectively, represented by primitive positive roots. (iii) For  $B = 19$ , the sharp lower bound for  $\eta = |A + P - 2E|/|D|$  is 347984603/9837828000, attained by exactly one class, already present for  $B = 13$ . These are unbounded-exponent statements: completeness uses Theorem A of von Känel–Matschke, not a search cutoff.*

## The four-prime frontier

For exact support  $\text{supp}(2d) = \{2, 3, p, q\}$  the paper proves the complete endpoint-incidence classification and edge identity  $F_{10} = F_{02}$ , the balanced-branch reduction to seven fixed primitive

cores, and the complete exclusion of the outer double-balanced core-area-6 branch. Aebi-core central genus-two fibres, outer fixed cores and persistent unbalanced cores remain open beyond the finite-support exclusions of Theorem 8; their residual states are in Section 9.

## What this paper does not claim

No existence or nonexistence of a  $3 \times 3$  magic square of nine distinct positive squares in general; no global Parker impossibility; no effective enumeration in an unrestricted squareclass; no arbitrary four-prime exclusion or support-size bound; no exact ranks or point lists beyond the certified ones; no decision of any object in the companion dossiers (the residual cover  $q_R(K)$ , the  $n \geq 5$  lift tower, the degree-90 class  $[A_R]$  — all open there); and **no novelty, priority or firstness anywhere**: prior-art gates for such wording are deliberately closed, and every no-hit search is reported as “no antecedent located, firstness unproved”.

## Detailed plan of the paper

- §1 Introduction, historical sources and entry-side versus difference-side questions.
- §2 The seven-line normal form, retraction bookkeeping and integral line-sum lattice.
- §3 The three infinite  $(8, 7)$  constructions, least-shift law and elliptic-orbit density.
- §4 The interaction surface, saturation and  $S_5$  cone map.
- §5 Imported finiteness, the closed centre-841 fibre, fixed Bremner shadow and finite Kummer-class stop.
- §6 The five-term  $S$ -unit law, valuation restrictions and  $\{2, 3\}$  exclusion.
- §7 Three-prime uniqueness from Aebi’s classification and the alternative endpoint proof for primitive configurations.
- §8 Complete small-support catalogues: the imported  $S$ -unit total, triangle correspondence, exclusion,  $(9, 7)$  atlas and sharp defect ratio.
- §9 The arbitrary four-prime frontier and its residual open branches.
- §10 The computational program and the three open research dossiers.
  - A The bounded  $(8, 7)$  census and catalogue provenance.
  - B The theorem-to-certificate map and frozen external-CAS records.
  - C Reproducibility, source dependencies, data policy and AI assistance.

# 1 Introduction

## 1.1 The problem and its records

Whether a  $3 \times 3$  magic square whose nine entries are distinct positive integer squares exists was posed by LaBar in 1984 and popularised by Gardner, who offered \$100 for an example; the problem also appears in Guy’s *Unsolved Problems in Number Theory* (this history is recorded in Bremner [Bre99], §0). It remains open in both directions. The two record objects are dual, and both have stood since 1997: a fully magic square with *seven* of nine entries square, due to Bremner [Bre99], and Sallows’s array of *nine* distinct squares with seven of its eight lines sharing one sum [Sal97]. Bremner states in the same place that an example with eight distinct square entries is unknown. Euler had already produced an order-4 magic square of distinct squares in 1770, so order three is exactly where the problem lives.

Matt Parker later made one celebrated near miss, now known as the Parker Square [Par]. The terminology “Parker configuration” used below recalls that public connection with the problem; it does not suggest that Parker posed the problem, and it does not identify his particular square with the common-difference configurations defined in this paper.

## 1.2 A rationality filter

A fully magic square of nine distinct squares *does exist* over a number field of degree four. Bremner displayed the underlying magic square with eight entries square in  $\mathbb{Q}(\sqrt{3})$  and centre 532, which is not a square in that field [Bre01]; adjoining  $\sqrt{133}$  makes the centre  $532 = (2\sqrt{133})^2$ . Thus the degree-four completion is the elementary field extension of Bremner’s displayed object, not an additional construction attributed to him. The companion repository replays it as an exact verified witness. Bremner [Bre99], §3, separately constructs an example over a field of degree 27. The consequence deserves emphasis: there is no combinatorial or shape-only obstruction to the Parker problem. The only property that can fail is rationality over  $\mathbb{Q}$ , so any impossibility argument must use an invariant that separates  $\mathbb{Q}$  from a quartic field. This filter shapes the whole paper: the positive constructions of Section 3 show how close the rational story comes, and the support laws of Sections 6–7 are genuinely rational-arithmetic statements, invisible over a larger field.

## 1.3 The entry side: what is known about a hypothetical solution

A hypothetical rational solution is constrained through the prime factors of its *entries*; throughout this paragraph the square is taken in the primitive normalization (unit gcd of the nine entries). Rabern [Rab03] proved — Weisenberg’s restatement in [Wei23], §2.1, matches the primary on all six results — that every prime factor of an entry is odd; that a prime dividing the centre is  $1 \pmod{4}$ ; that a prime  $p \equiv 3, 5 \pmod{8}$  dividing a non-centre entry also divides the centre and the third entry of that line; that no  $p \equiv 3 \pmod{8}$  divides any entry; that no  $p \equiv 5 \pmod{8}$  divides a middle-side entry; and that a  $p \equiv 3 \pmod{4}$  dividing a corner divides the two non-adjacent middle-side entries. Woll [Wol18] proved that a prime  $p$  dividing only the centre forces  $p$  to possess three consecutive nonzero quadratic residues, and counted the resulting residue classes; Labruna [Lab18] studied the order-three analogue over finite fields, where in characteristic greater than three a nontrivial magic square of squares attains exactly 3, 5, 7 or 9 distinct entries, each degree realized over infinitely many prime fields. Weisenberg [Wei23] added the order bookkeeping (a prime dividing exactly three entries attains its minimum order at least twice), the restriction  $p \equiv 1 \pmod{24}$  for a prime dividing exactly two entries, consecutive-residue patterns for single-entry primes, a

complete seven-arrangement list of the ways a prime factor can appear, and a proposal — not executed there — to organize searches by the Gaussian prime signature of the centre. A recent preprint of Hill [Hil26] announces a nonexistence proof; it is unrefereed, and nothing in this paper depends on it. Coumbe [Cou24] develops a Pell-transform approach and claims that the smallest entry cannot be unity or a prime square. We do not import that exclusion: the unrestricted difference-set statement used in its proof has the literal instance  $120 + 120 = 240$ , coming from the two primitive progressions  $(7^2, 13^2, 17^2)$  and  $(7^2, 17^2, 23^2)$ . This does not refute a repaired distinct-summand statement or the final exclusion itself, but it prevents the published proof from being used here without repair.

All of these statements constrain *entries of a completed square*. The present paper works on a different quantity — the common difference of the three square progressions forced by the completion problem — and Section 6 depends on none of the entry-side results; the two layers meet only in the classical toolbox of quadratic residues and sums of two squares.

## 1.4 The difference side: what this paper proves

The objects and the eight main theorems are stated in full in the front matter. In one paragraph: every seven-line array is three same-difference arithmetic progressions and its integral line-sum cokernel is  $\mathbb{Z} \oplus \mathbb{Z}/3\mathbb{Z}$  (Theorem 1 and Proposition 1); three explicit mechanisms produce infinite  $(8, 7)$  families, with a complete least-shift law for the first and a positive-density orbit law for the third (Theorem 2); completing the eighth line of the transversal frame is a single equation on  $E_{840}^3$ , whose saturated model is one irreducible surface with an  $S_5$  cone (Theorem 3); the recent preprint of Harrison–Mudgal–Schmidt [HMS26] gives uniform rank-dependent bounds for progressions inside one elliptic coordinate set, a complementary result which does not decide this three-coordinate interaction. For fixed difference the relevant points are finite by a theorem imported from Caro and García-Fritz [CGF25], one fibre is closed completely, and finite Kummer classes alone can never decide the interaction (Theorems 4 and 5); additionally, the rational shadow line extracted from Bremner’s degree-four example has no nonconstant all-rational-square member (Proposition 6); the common difference obeys a five-term  $S$ -unit law with an endpoint/interior valuation table (Theorem 6, resting on the subspace theorem of Evertse, Schlickewei and Schmidt and on the classical equal-area correspondence recorded by Robertson [Rob96]); Aebi’s classification [Aeb26] implies uniqueness of the square progression for every difference with at most three prime divisors (Theorem 7). Finally, the complete  $S$ -unit totals of von Känel–Matschke [vKM16], followed by an exact finite reconstruction, exclude full-magic interactions for differences supported on the primes at most 19 and classify 384  $(9, 7)$  classes with a sharp difference-normalized defect bound (Theorem 8). The arbitrary four-prime problem remains at the explicit frontier of Section 9.

## 1.5 Method, certificates, and what is not claimed

The companion repository supplies exact Python replays for the elementary and symbolic computations. Results depending on external computer algebra — ranks, complete point lists and Selmer sets — ship instead as hash-pinned inputs and normalized output records: the public verifier authenticates and parses those records but does not independently recompute the underlying calculations. Bounded censuses corroborate statements but never prove an unbounded one. Catalogued web data (Boyer’s pages [Boy05, Boy], Fituvalu’s tables, Brown’s centre records) is used under explicit provenance locks, with the corrections recorded in Appendix A. The what-this-paper-does-not-claim block of the front matter is part of the paper’s contract: in particular, no novelty or priority is asserted for any statement, and every no-hit prior-art search is reported as

exactly that.

## 2 The seven-line normal form

Houston [Hou16a, Hou16b], starting from the classical Lucas parametrization, explained this almost-magic normal form and its interpretation as three three-term arithmetic progressions of one common difference. We retain a self-contained proof because the integral image lattice below is needed to identify the exact congruence obstruction.

**Lemma 1** (Retraction bookkeeping). *Let a fully magic  $3 \times 3$  array have common line sum  $M$ . (i) Subtracting the same  $t$  from the three cells of one long diagonal leaves the three rows, the three columns and the other diagonal with sum  $M - t$ , while the selected diagonal has sum  $M - 3t$ ; the exact line defect is  $2t$ . (ii) More generally, subtracting  $\tau$  from the three cells of a permutation transversal  $T$  leaves every row and every column with sum  $M - \tau$ ; if  $T$  meets the two diagonals in  $j$  and  $k$  cells, their sums become  $M - j\tau$  and  $M - k\tau$ . (iii) Conversely, if exactly one diagonal of a  $3 \times 3$  array fails while the other seven lines share a common sum, adding half the defect to each cell of the failed diagonal restores a fully magic array.*

*Proof.* (i) Every row and every column contains exactly one modified cell, and the other diagonal contains exactly one modified cell, while the selected diagonal contains all three; the sums follow, and the defect is  $(M - t) - (M - 3t) = 2t$ . Nothing here uses squareness or any property of a particular family. (ii) A permutation transversal meets every row and every column in exactly one cell, and meets the diagonals in  $j$  resp.  $k$  cells by definition of  $j, k$ ; the same count gives the sums. (iii) By Theorem 1 (proved independently below) the array is the displayed normal form, with seven-line sum  $A + E + P$ , failed antidiagonal sum  $3E$ , and defect  $\delta = A + P - 2E$ . Each row and column contains exactly one antidiagonal cell, and the principal diagonal contains one (the centre), so adding  $\delta/2$  to the three antidiagonal cells raises all seven line sums to  $A + E + P + \delta/2$  and the antidiagonal to  $3E + 3\delta/2$ ; both equal  $\frac{3}{2}(A + P)$ , so all eight lines agree.  $\square$

**Proposition 1** (Integral line-sum lattice). *Let  $\Lambda : \mathbb{Z}^9 \rightarrow \mathbb{Z}^8$  send  $X = (a, b, c; d, e, f; g, h, i)$  to  $(R_1, R_2, R_3, C_1, C_2, C_3, D_+, D_-)$ , its three row sums, three column sums and two diagonal sums. Then*

$$\text{SNF}(\Lambda) = \text{diag}(1, 1, 1, 1, 1, 1, 3, 0), \quad \text{coker}(\Lambda) \cong \mathbb{Z} \oplus \mathbb{Z}/3\mathbb{Z}.$$

*An integer vector of eight prescribed line sums lies in the image if and only if*

$$R_1 + R_2 + R_3 = C_1 + C_2 + C_3, \quad R_1 + R_3 \equiv C_2 + D_+ + D_- \pmod{3}.$$

*Consequently  $\Lambda$  has rank 6 over  $\mathbb{F}_3$  and rank 7 over  $\mathbb{F}_p$  for every prime  $p \neq 3$ .*

*Proof.* In the entry order  $(a, b, c, d, e, f, g, h, i)$ , the matrix of  $\Lambda$  is

$$\begin{pmatrix} 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 0 \end{pmatrix}.$$

The primitive left relation  $-R_1 - R_2 - R_3 + C_1 + C_2 + C_3 = 0$  bounds the rank by 7. The minor on rows  $(R_1, R_2, R_3, C_1, C_2, D_+, D_-)$  and columns  $(a, b, c, d, e, f, g)$  is  $-3$ , whereas the minor on rows  $(R_1, R_2, R_3, C_1, C_2, D_+)$  and columns  $(a, b, c, d, f, g)$  is  $-1$ . Moreover

$$R_1 + R_3 - C_2 - D_+ - D_- = -3e,$$

so modulo 3 the rank is at most 6 and every 7-minor is divisible by 3. The sixth and seventh determinantal divisors are therefore 1 and 3, giving the displayed Smith form. The two stated conditions annihilate the image and define a surjective map  $\mathbb{Z}^8 \rightarrow \mathbb{Z} \oplus \mathbb{Z}/3\mathbb{Z}$  (use the  $R_2$  and  $D_+$  coordinate vectors); its kernel has the same quotient as  $\text{im } \Lambda$ , so the two coincide. Reduction of the Smith form gives the rank statement.  $\square$

*Proof of Theorem 1.* The failed line cannot be a row: the three columns already force the total  $3S$ , so if two rows have sum  $S$  the third does as well. The same argument excludes a failed column. Up to the dihedral action of the grid the failed line is therefore the antidiagonal. Proposition 1 forces its sum to be divisible by 3. Solving the seven linear line equations then shows that its central cell is one third of that sum and gives the displayed universal form; the seven common lines have sum  $A + E + P$  and the antidiagonal has sum  $3E$ . The nine entries decompose into the three arithmetic progressions

$$(A - D, A, A + D), \quad (P - D, P, P + D), \quad (E - D, E, E + D),$$

of one common difference  $D$ . Consequently an array with exactly eight square entries consists of two full three-square progressions and one progression containing exactly two squares, which is the last assertion.  $\square$

*Remark 1* (Where full magic lives). In the normal form, the array is fully magic if and only if the antidiagonal sum equals the common sum, that is  $3E = A + E + P$ , equivalently

$$A + P = 2E :$$

the three centres themselves must form a three-term arithmetic progression. For an all-squares array this is a *fourth* square progression, with its own common difference  $E - A$  — never equal to  $\pm D$ , since  $E = A + D$  would repeat the centre at positions  $(2, 2)$  and  $(2, 3)$ , while  $E = A - D$  would repeat it at  $(2, 2)$  and  $(3, 2)$ , against nine-way distinctness — and this single extra condition is exactly where the elliptic wall of Section 4 lives. The classical one-parameter families of nine-square arrays with seven equal sums (Lucas [Luc76]; Sallows's record example [Sal97]) sit inside this normal form, as does the general parametrization of  $3 \times 3$  magic squares going back to Lucas [Luc91].

*Remark 2* (Census normalization). The bounded census of Appendix A quotients by the geometric  $D_4$  action and measures height by the largest square root among the eight square entries; its exact minimum statements are supporting propositions at their frozen bound and are never used as evidence for an unbounded claim.

### 3 Infinite $(8, 7)$ families

This section proves the four parts of Theorem 2. Three mechanisms are involved: a diagonal retraction of a catalogued six-square carrier family, a bridge completion inside a catalogued carrier type, and an elliptic-orbit lift of one fixed carrier.

### 3.1 The retraction family and its least-shift law

The carrier is the six-square fully magic family recorded on Boyer's catalogue pages [Boy] and attributed there to Wesolowski; the catalogue states explicitly that the recurrence below "was not given by Arkadiusz", and the  $n = 1$  grid already occurs as Configuration II of Bremner [Bre01], so we speak of the Bremner/Wesolowski carrier. With  $(x_0, y_0, z_0) = (3, 1, 1)$  and

$$x_{n+1} = 3x_n + 2z_n, \quad y_{n+1} = x_n, \quad z_{n+1} = x_n + z_n,$$

the carrier  $W_n$  is fully magic with centre  $E = 4z^4 + 5z^2 + 1$  and common sum  $3E$ , where  $(x, y, z) = (x_n, y_n, z_n)$ . Subtracting  $t = 6z^2$  from its antidiagonal and using the recurrence identities

$$x^2 + y^2 z^2 - 6z^2 = (yz + 1)^2, \quad x^2 z^2 + y^2 - 6z^2 = (xz - 1)^2, \quad \frac{(x^2 + y^2)(z^2 + 1)}{2} - 6z^2 = 4z^4 - z^2 + 1$$

produces

$$N_n = \begin{pmatrix} (xy - z)^2 & (xz + y)^2 & (yz + 1)^2 \\ (x + yz)^2 & 4z^4 - z^2 + 1 & (xz - y)^2 \\ (xz - 1)^2 & (yz - x)^2 & (xy + z)^2 \end{pmatrix}.$$

*Proof of Theorem 2(i).* By Lemma 1(i) the seven unselected lines of  $N_n$  share the sum  $3E - 6z^2 = 3(4z^4 + 3z^2 + 1)$ , the antidiagonal sums to  $3E - 18z^2 = 3(4z^4 - z^2 + 1)$ , and the defect is  $12z^2$ . For  $z > 1$ ,

$$(2z^2 - 1)^2 < 4z^4 - z^2 + 1 < (2z^2)^2,$$

so the centre is not a square. The eight positive roots are, in increasing order,

$$yz - x < yz + 1 < x + yz < xy - z < xy + z < xz - y < xz - 1 < xz + y :$$

positivity of the first follows from  $yz - x = y(z - 1) - 2z \geq 3(z - 1) - 2z = z - 3 > 0$ , most gaps are immediate, and the two non-obvious ones reduce, through the recurrence invariants, to

$$(xy - z) - (x + yz) = 2z^2 - yz - y - 3z + 1 > 0, \quad (xz - y) - (xy + z) = (y - 1)(z - 1) - 2 > 0,$$

the first bounded below by  $(z - 1)(z - 2) > 0$  using  $y \leq z - 1$ , the second at least 4. Hence all nine entries are positive and distinct. For root-primitivity: an integer dividing all eight roots divides both  $(yz + 1) - (yz - x) = x + 1$  and  $(x + yz) - (yz + 1) = x - 1$ , hence divides 2; the recurrence keeps  $x, y$  odd and toggles the parity of  $z$ , so for odd  $n$  all eight roots are odd and their gcd is 1. There are infinitely many odd indices.  $\square$

**Proposition 2** (Complete pair-closing classification). *Fix  $n$  and write the retracted antidiagonal entries of the carrier as  $C < E < G$  with common gap  $\Delta$ . Every positive shift  $0 < t < C$  for which at least two of  $C - t$ ,  $E - t$ ,  $G - t$  are squares arises from a same-parity factor pair, and every retained pair works: for branch  $CE$ , from  $de = \Delta$ ,  $0 < d < e$ ,  $d \equiv e \pmod{2}$ , via  $u = (e - d)/2$ ,  $w = (e + d)/2$ ,  $t = C - u^2$ ; for branch  $EG$ , from  $de = \Delta$  via  $w = (e - d)/2$ ,  $v = (e + d)/2$ ,  $t = E - w^2$ ; for branch  $CG$  from  $de = 2\Delta$  via  $u = (e - d)/2$ ,  $v = (e + d)/2$ ,  $t = C - u^2$ .*

*Proof.* For  $CE$ , subtracting the two square equations gives  $w^2 - u^2 = \Delta = (w - u)(w + u)$ , so  $d = w - u$  and  $e = w + u$  have the stated product and parity, and the inverse formulas recover  $u, w$ . The other branches are identical with difference  $\Delta$  or  $2\Delta$ . This is a divisor classification, not a scan over  $t$ .  $\square$

**Proposition 3** (Least shift). *For every  $n \geq 2$ ,  $t_0 = 6z^2$  is the least positive shift closing at least one pair; at  $n = 1$  the least is 9. Hence  $t_{\min}(1) = 9$  and  $t_{\min}(n) = 6z_n^2$  for  $n \geq 2$ . Leastness does not mean uniqueness: larger nondegenerate closing shifts exist and are recorded exactly.*

*Proof frame.* For  $n \geq 2$  the recurrence on  $(y, z)$  is  $(y, z) \mapsto (y + 2z, y + 3z)$ , starting at  $(11, 15)$ , and direct induction gives

$$z > y, \quad 2y > z, \quad 5y > 3z, \quad 5z > 6y, \quad 8y > 5z, \quad z \geq 15.$$

Assume  $0 < t < t_0$  closes a pair and put  $h = t_0 - t$ ,  $0 < h < 6z^2$ . At  $t_0$  the three entries are  $C - t_0 = Y^2$  with  $Y = yz + 1$ ,  $E - t_0 = 4z^4 - z^2 + 1$ , and  $G - t_0 = X^2$  with  $X = xz - 1$ ; each entry can be square inside the short interval only along an explicit one-parameter list (for the  $C$  entry,  $C - t = (Y + a)^2$  forces  $h = a(2Y + a)$ , and similarly for  $E$  and  $G$ ), and the listed windows are excluded pairwise by the induction inequalities. The complete three-case interval analysis, with its exact certificate, is reproduced in Appendix B. At  $n = 1$ ,  $(C, E, G) = (265, 1105, 1945)$ : for  $1 \leq t \leq 8$  each retracted entry lies strictly between consecutive squares ( $16^2 < C - t < 17^2$ ,  $33^2 < E - t < 34^2$ ,  $44^2 < G - t < 45^2$ ), so no pair closes below  $t = 9$ , while  $t = 9$  gives  $(256, 1096, 1936) = (16^2, 1096, 44^2)$ .  $\square$

Parts (i)–(ii) of Theorem 2 are proved.

### 3.2 The bridge completion in type 6.VI

The catalogued carrier type 6:6 (6.VI here) is normalized, after the dihedral action, as

$$M(A, N, Z) = \begin{pmatrix} 2N - Z & A & N - A + Z \\ 2Z - A & N & 2N - 2Z + A \\ N + A - Z & 2N - A & Z \end{pmatrix},$$

with square cells  $A = a^2$ ,  $N = n^2$ ,  $Z = z^2$ ,  $B = 2N - A = b^2$ ,  $D = 2Z - A = d^2$ ,  $C = 2N - D = c^2$ . The type taxonomy and database records are Fituvalu's [Fit]; the fixed numerical carrier used below is historically Bremner's Configuration VI [Bre01].

**Proposition 4** (The four-progression network). *The six source squares satisfy  $a^2 + b^2 = 2n^2$ ,  $d^2 + c^2 = 2n^2$  and  $d^2 + a^2 = 2z^2$ , so the source contains the three square progressions  $(a^2, n^2, b^2)$ ,  $(d^2, n^2, c^2)$ ,  $(d^2, z^2, a^2)$ . With  $\delta = a^2 - z^2 = z^2 - d^2 > 0$ , the selected antidiagonal is identically  $(n^2 - \delta, n^2, n^2 + \delta)$ , and subtracting  $t > 0$  closes it to squares  $(x^2, y^2, w^2)$  exactly when*

$$x^2 = y^2 - \delta, \quad w^2 = y^2 + \delta, \quad t = n^2 - y^2 > 0 :$$

*a fourth square progression, with the difference of the bridge  $(d^2, z^2, a^2)$ .*

*Proof.* Substituting  $D = 2Z - A$  into the normalized array gives  $A - Z = Z - D = \delta$ , and the antidiagonal entries become  $N - \delta$ ,  $N$ ,  $N + \delta$ ; a common subtraction preserves their differences, and the closure conditions are the displayed ones.  $\square$

**Proposition 5** (Complete bridge-completion classification). *Fix an increasing bridge progression  $(d^2, z^2, a^2)$  of difference  $\delta$ . Every positive type-6.VI completion arises from a same-parity factor pair*

$$uv = 2\delta, \quad 0 < u < v, \quad u \equiv v \pmod{2},$$

via  $b = (v - u)/2$ ,  $c = (v + u)/2$ ,  $n^2 = (a^2 + b^2)/2$ , a pair being retained exactly when  $n^2$  is an integer square; then  $d^2 + c^2 = a^2 + b^2 = 2n^2$  automatically. The pair  $(u, v) = (a - d, a + d)$  always returns the tautological completion  $(b, n, c) = (d, z, a)$ , which repeats square cells and is not admissible. Positivity, nine-way distinctness and the six-square mask are separate final checks.

*Proof.* Subtracting the two centre relations gives  $c^2 - b^2 = a^2 - d^2 = 2\delta$ , so  $u = c - b$ ,  $v = c + b$  have the stated product and parity, and inversion recovers  $b, c$ ; conversely any retained pair satisfies  $c^2 - b^2 = 2\delta = a^2 - d^2$ , so the two expressions for  $2n^2$  agree.  $\square$

*Proof of Theorem 2(iii).* For the fixed bridge  $(23^2, 37^2, 47^2)$ ,  $\delta = 840$ : running Proposition 5 over the finitely many same-parity factor pairs of  $2\delta = 1680$  retains exactly the tautological  $(23, 37, 47)$  and the unique admissible completion  $(79, 65, 89)$  (exact finite check; certificate in Appendix B). Coupling the resulting carrier with Robertson's classical progression  $(1^2, 29^2, 41^2)$  of the same difference [Rob96] retracts the antidiagonal by  $\tau = 65^2 - 29^2 = 3384$  and yields the explicit  $(8, 7)$  array

$$\begin{pmatrix} 7081 & 2209 & 1 \\ 529 & 841 & 7921 \\ 1681 & 6241 & 1369 \end{pmatrix}$$

with seven-line sum 9291, failed antidiagonal sum 2523, and unique nonsquare 7081.  $\square$

### 3.3 The elliptic-orbit lift

The fixed carrier is

$$M_0 = \begin{pmatrix} 7081 & 2209 & 3385 \\ 529 & 4225 & 7921 \\ 5065 & 6241 & 1369 \end{pmatrix},$$

Bremner's Configuration VI [Bre01], in the normalized orientation of Fituvalu database row 340545 [Fit], is fully magic with sum 12675. Its six square entries have outside roots

$$\{23, 37, 47, 79, 89\},$$

and its selected antidiagonal is  $(65^2 - 840, 65^2, 65^2 + 840)$ . Let

$$E: Y^2 = X^3 - 840^2 X, \quad Q = (1960, 78400), \quad P = 2Q = (841, -1189).$$

**Lemma 2** (Doubling produces a square progression). *For  $R = (u, v) \in E(\mathbb{Q})$  with  $v \neq 0$ ,*

$$x(2R) = \left( \frac{u^2 + 840^2}{2v} \right)^2, \quad x(2R) \mp 840 = \left( \frac{u^2 \mp 1680u - 840^2}{2v} \right)^2,$$

*so every finite point of  $2E(\mathbb{Q})$  supplies three rational squares in arithmetic progression with common difference 840 — the classical congruent-number correspondence (cf. [Rob96]), in the exact convention used by the lift.*

**Lemma 3** (Fixed-carrier lift). *Let  $R \in 2E(\mathbb{Q})$  with  $840 < x(R) < 65^2$ . Write the positive roots of  $x(R) - 840$ ,  $x(R)$ ,  $x(R) + 840$  in lowest terms and let  $\lambda$  be the least common multiple of their denominators. Scale  $M_0$  by  $\lambda^2$  and subtract  $\tau = \lambda^2(65^2 - x(R))$  from the selected antidiagonal; the antidiagonal becomes the three cleared squares. If the three new roots avoid  $\{23, 37, 47, 79, 89\}$ , the output is positive with nine distinct entries and exactly eight squares, the unique nonsquare being  $7081\lambda^2$  — and it is entry-primitive: a common prime  $p$  of all entries would divide  $\lambda$ ; choosing a reduced root denominator of maximal  $p$ -adic valuation, its cleared numerator stays prime to  $p$ , and its square blocks  $p$ . The argument uses minimality of  $\lambda$  and fails for an arbitrary common multiple.*

**Lemma 4** ( $Q$  is non-torsion).  $P = 2Q = (841, -1189)$  is integral with  $1189 = 29 \cdot 41$ , while  $4 \cdot 840^6$  has only the prime divisors  $2, 3, 5, 7$ ; by the Lutz–Nagell criterion [Sil09, Chap. VII, Prop. 3.1] an integral torsion point with nonzero ordinate would satisfy  $Y^2 \mid 4 \cdot 840^6$ , so  $P$  is non-torsion, hence so is  $Q$ . Independently,  $\#E(\mathbb{F}_{11}) = 12$  and  $\#E(\mathbb{F}_{13}) = 20$  at good primes bound the rational torsion order by  $\gcd(12, 20) = 4$ , and  $P$  is neither infinity nor rational two-torsion.

*Proof of Theorem 2(iv).*  $P$  lies in the identity component  $E(\mathbb{R})^0$ , a circle group, and is non-torsion, so its positive multiples are dense there. The open set  $U = \{R \in E(\mathbb{R})^0 : 840 < x(R) < 4225\}$  contains  $P$ ; removing the finite set where one of the three roots equals one of  $23, 37, 47, 79, 89$  leaves a nonempty open set  $V$  — the roots at  $P$  itself are  $(1, 29, 41)$ . Density gives infinitely many positive multiples  $mP = 2mQ$  in  $V$ , and Lemma 3, always with the least common multiple, turns each return into a primitive positive  $(8, 7)$  array. If the outputs at indices  $m, n$  were related by a  $D_4$  action and a positive square scale  $\rho^2$ , the unique nonsquare entries would force  $7081\lambda_n^2 = \rho^2 7081\lambda_m^2$ , so  $\rho = \lambda_n/\lambda_m$ ; a  $D_4$  action fixes the centre, so comparing centres gives  $x(mP) = x(nP)$ , hence  $mP = \pm nP$ , and non-torsion with  $m, n > 0$  forces  $m = n$ . Distinct return indices therefore give distinct square-scale/ $D_4$  classes. Two exact witnesses, including the  $m = 3$  return, are frozen in the companion repository.  $\square$

**Corollary 1** (Positive-density orbit lift). *Let  $A(N)$  count the indices  $1 \leq m \leq N$  for which  $mP$  lies in the admissible set  $V$  of the preceding proof. Then*

$$A(N) = \delta N + o(N), \quad \delta = \frac{\int_{840}^{4225} dx/\sqrt{x^3 - 840^2 x}}{\int_{840}^{\infty} dx/\sqrt{x^3 - 840^2 x}} = 1 - I_{(840/4225)^2} \left( \frac{1}{4}, \frac{1}{2} \right),$$

and

$$\delta = 0.6585271498271213 \dots$$

*The same asymptotic counts the square-scale/ $D_4$ -inequivalent primitive  $(8, 7)$  classes produced among  $P, 2P, \dots, NP$ .*

*Proof.* The multiples of the non-torsion point  $P$  are equidistributed in the circle group  $E(\mathbb{R})^0$  with respect to normalized Haar measure; see Cowan [Cow20] for the natural-density formulation for real elliptic curves. The boundary of  $840 < x < 4225$  has measure zero, and deleting the finitely many collision points changes neither measure nor density. On the identity component the invariant measure is proportional to  $dx/\sqrt{x^3 - 840^2 x}$ , giving the ratio above. Substitution  $x = 840/t^2$  changes it to

$$\frac{\int_{\sqrt{840/4225}}^1 dt/\sqrt{1-t^4}}{\int_0^1 dt/\sqrt{1-t^4}},$$

which is the stated regularized incomplete-beta value. The final counting claim follows from the inequivalence argument in the preceding proof.  $\square$

**Remark 3** (The universal lemniscatic law behind  $\delta$ ). The special-function value in Corollary 1 is not peculiar to 840. For  $d > 0$  let

$$E_d : \quad Y^2 = X^3 - d^2 X.$$

If  $R = (X, Y)$  is distributed according to normalized Haar probability on  $E_d(\mathbb{R})^0$  and  $U = (d/X)^2$ , then the push-forward density is

$$d\mu_U(u) = \frac{u^{-3/4}(1-u)^{-1/2}}{B(\frac{1}{4}, \frac{1}{2})} du, \quad 0 < u < 1;$$

equivalently,  $U \sim \text{Beta}(\frac{1}{4}, \frac{1}{2})$ . Indeed,  $X = du^{-1/2}$  sends the invariant differential  $dX/\sqrt{X^3 - d^2X}$  to a constant multiple of  $u^{-3/4}(1-u)^{-1/2}du$ . Consequently, for every  $B > d$ ,

$$\mu\{R \in E_d(\mathbb{R})^0 : d < X(R) < B\} = 1 - I_{(d/B)^2}\left(\frac{1}{4}, \frac{1}{2}\right).$$

The normalizing integral is the classical lemniscatic value

$$\int_0^1 \frac{dt}{\sqrt{1-t^4}} = \frac{\Gamma(1/4)^2}{4\sqrt{2\pi}}$$

recorded in the NIST Digital Library of Mathematical Functions [DLMF, §19.20(i), Eq. (19.20.2)]. This universality reflects  $j(E_d) = 1728$ : after real scaling, every curve in the family has the same normalized invariant measure, so the probability depends only on  $B/d$ . For a non-torsion point in  $E_d(\mathbb{R})^0$ , equidistribution transfers this probability to orbit indices. A point on the other real component requires separate bookkeeping for its even and odd multiples. Thus the remark explains the density constant but adds no new arithmetic input to the corollary.

*Remark 4* (From constructions to the wall). Part (iv) reaches the  $(8, 7)$  metric infinitely often from one carrier. Closing the eighth line of the same transversal frame is the interaction problem of Section 4: by Remark 1 it demands a fourth square progression through the centres, and that single extra condition changes the landscape from constructive to arithmetic.

## 4 The interaction surface

### 4.1 The transversal equation

For arbitrary  $x_0, x_1, x_2$  consider

$$\begin{pmatrix} x_1 & x_2 + 840 & x_0 - 840 \\ x_2 - 840 & x_0 & x_1 + 840 \\ x_0 + 840 & x_1 - 840 & x_2 \end{pmatrix}.$$

Every row, every column and the principal diagonal sums to  $x_0 + x_1 + x_2$ , while the antidiagonal sums to  $3x_0$ ; all eight sums agree if and only if

$$x_1 + x_2 = 2x_0.$$

This proves Theorem 3(i): imposing that each  $x_i$  be the  $x$ -coordinate of a point of  $2E_{840}(\mathbb{Q})$  — the arithmetic condition of Lemma 2 — makes the fully magic completion of the transversal frame exactly one linear equation on  $E_{840}^3$ , hence a surface. The diagonal  $(P, P, P)$  is not isolated, and the membership condition  $P_i \in 2E_{840}(\mathbb{Q})$  is arithmetic rather than a further geometric equation over an algebraic closure.

For later use we record the Mordell–Weil source lock. The minimal model of  $E_{840}$  is recorded as LMFDB 705600.vn3 [LMFDB],  $Y^2 = X^3 - 44100X$ , with rank 2, full rational 2-torsion and

free generators  $(-84, 1764)$ ,  $(294, 3528)$ ; transporting by  $(x, y) = (4X, 8Y)$  and adjusting by the two-torsion point  $T = (-840, 0)$  gives the free basis

$$Q = (1960, 78400), \quad R = (1176, 28224)$$

used throughout, with

$$2Q = (841, -1189), \quad 2R = (1369, -39997).$$

Since doubling kills the rational two-torsion,  $2Q$  and  $2R$  generate the free part of  $2E_{840}(\mathbb{Q})$ .

## 4.2 Saturation: one surface, not a bundle of components

Normalizing the three same-difference square progressions (the weave coordinates of the companion documents) reduces the Parker open locus to the affine variety  $Y$  cut out by

$$\begin{aligned} F &= 2R^2S^2 - (1 + W^2)(R^2 + S^2) + 2W^2 = 0, \\ A^2 &= W^2 + R^2S^2, & B^2 &= W^2R^2 + S^2, & C^2 &= W^2S^2 + R^2. \end{aligned}$$

with the Parker open  $Y^\circ$  removing  $RSWABC = 0$ , the nine zero-cell divisors and the 27 cross-progression collision divisors (30 distinct normalized factors in all); positivity stays a real semialgebraic condition and is never encoded as a polynomial saturation.

*Proof of Theorem 3(ii), saturation part.* As a polynomial in  $S$ ,

$$F = (2R^2 - W^2 - 1)S^2 + 2W^2 - (1 + W^2)R^2,$$

whose leading coefficient and constant term are coprime; the coefficient  $2R^2 - W^2 - 1$  is squarefree and occurs to odd order in the ratio defining  $S^2$ , so that ratio is not a square in  $\mathbb{Q}(R, W)$  and  $F$  is irreducible. Over the function field of  $F = 0$  put  $p_A = W^2 + R^2S^2$ ,  $p_B = W^2R^2 + S^2$ ,  $p_C = W^2S^2 + R^2$ . In squared coordinates  $r = R^2$ ,  $s = S^2$ ,  $w = W^2$ , setting  $p_A = 0$  gives  $F(r, s, -rs) = (r + s)(rs - 1)$ , and on the reduced component  $RS = 1$ ,  $W^2 = -1$  the function  $p_A$  has generic valuation one while  $p_B$  and  $p_C$  are generically nonzero; the case  $p_B = 0$  reduces to  $-2r^2w + rw^2 - r + 2w = 0$ , linear in  $p_B$  at the generic point of each reduced component with  $p_A$  and  $p_C = 2p_A$  generically nonzero; the  $p_C$  case is symmetric. Hence every nonempty product of the three radicands has odd valuation somewhere and is not a square, so adjoining  $A, B, C$  gives a degree-eight extension with integral coordinate ring: the four equations define a single prime component of dimension two. None of the declared zero/collision polynomials vanishes on it, so saturating the prime ideal by their product changes nothing, and  $Y^\circ$  is one irreducible surface.  $\square$

## 4.3 The square-progression cone and its $S_5$ wall

The four equations imply the exact identity

$$2A^2 = B^2 + C^2 :$$

the three auxiliary roots themselves form a square progression, and the normalized interaction maps  $Y$  onto this classical cone. In squared coordinates  $a = rs + w$ ,  $b = rw + s$ ,  $c = sw + r$  (so  $2a = b + c$ ), reconstruction away from  $r^2 = 1$  is

$$s = \frac{b - ar}{1 - r^2}, \quad w = \frac{a - br}{1 - r^2},$$

where  $r$  satisfies the quintic

$$Q(r) = c(1 - r^2)^2 - (b - ar)(a - br) - r(1 - r^2)^2 = 0.$$

*Proof of Theorem 3(ii), cone part.* The quintic has leading coefficient  $-1$  and the displayed inverse reconstructs  $s, w$  from each generic root, so the squared-coordinate map to the cone is generically finite of degree five; each squared triple has eight sign choices of  $(R, S, W)$ , giving degree  $5 \cdot 8 = 40$  on the full signed cover. For the monodromy, specialize at the rational cone point  $(A, B, C) = (5, 1, 7)$ : the reconstruction polynomial

$$-r^5 + 49r^4 + 2r^3 - 123r^2 + 625r + 24$$

is irreducible modulo 7, factors modulo 29 with degree pattern  $[2, 1, 1, 1]$ , and has nonzero discriminant  $-24364389070416096000$ , so the specialization is separable. Irreducibility supplies a transitive degree-five group containing a five-cycle; the mod-29 pattern supplies a transposition; among transitive subgroups of  $S_5$  this forces the full  $S_5$ , and the generic monodromy contains the specialized one.  $\square$

*Remark 5* (What the wall does and does not say). Generic  $S_5$  monodromy proves the cone map has no generic rational section and that its quintic cannot be resolved by a tower of quadratic radicals. It does *not* prove that  $Y$  admits no other useful rational map, no height-decreasing correspondence, and no further structure; those remain open research directions, not consequences.

## 5 Finiteness, the closed fibre, and the finite-class stop

### 5.1 Fixed difference and fixed squareclass

*Proof of Theorem 4(i). Imported layer.* Caro and García-Fritz [CGF25] classify the images of nonconstant morphisms  $E \rightarrow X_{(c_1, c_2, c_3)} \subset E^3$  cut out by  $c_1x(P_1) + c_2x(P_2) + c_3x(P_3) = 0$  (their Theorem 1.2, four types) and bound, via Uniform Mordell–Lang, the intersection of the complement of those curves with any finite-rank subgroup of  $E^3$  (their Theorem 1.3). Both are imported; nothing below re-proves them.

**Exact specialization.** Over  $\mathbb{C}$  every  $E_d$  has  $j = 1728$ ,  $\text{End}(E_d) = \mathbb{Z}[i]$ , unit squares  $\pm 1$ ; take  $(c_0, c_1, c_2) = (-2, 1, 1)$ . Type (a) has two coordinates at infinity, so no finite Parker centre. In type (b) the cancellation  $c_i + u^2c_j = 0$  forces the moving pair to be the equal-coefficient pair  $(1, 1)$  with  $u^2 = -1$ , and the constant coordinate (the  $-2$  slot) satisfies, in the Weierstrass convention of [CGF25] with quadratic coefficient  $A = 0$ ,  $x(R) = -A(c_i + c_j)/(3c_k) = 0$ : the two-torsion point  $(0, 0)$ , whose square progression data is nonpositive or repeated, disjoint from the positive distinct locus. In type (c) the unit condition  $-2 + 1/u^2 + 1/v^2 = 0$  with inverse unit squares  $\pm 1$  forces  $u^2 = v^2 = 1$ , i.e. three equal  $x$ -coordinates — the repeated diagonal. Type (d) is absent twice over: its exceptional curve has CM by  $\mathbb{Z}[\sqrt{-2}]$  while  $E_d$  has CM by  $\mathbb{Z}[i]$ , and no permutation of  $(-2, 1, 1)$  satisfies  $c_1 = 2c_2 = 2c_3$ . Hence every classified curve inside  $X_d$  avoids the positive distinct Parker locus, and applying the imported Theorem 1.3 to  $\Gamma = \Gamma_d^3$  of rank  $3r$  gives finiteness with the stated  $D^{3r+1}$  shape. The bound is non-effective and yields no enumeration.

**Squareclass passage.** Writing  $d = q^2k$ , the isomorphism  $E_k \rightarrow E_d$ ,  $(X, Y) \mapsto (q^2X, q^3Y)$  commutes with doubling and scales  $x$ -coordinates, the difference and the interaction relation all by  $q^2$ , preserving positivity and distinctness; every solution with difference in the squareclass  $[k]$  descends to one of the finitely many on  $E_k$ .  $\square$

### 5.2 The fibre $x_0 = 841$ is closed

Here  $841 = x(2Q)$  is the centre of Robertson’s classical progression.

*Proof of Theorem 4(ii).* Fix  $x_0 = 841$ , write  $x_1 = 841+t$ ,  $x_2 = 841-t$ , and put  $f(x) = x(x^2-840^2)$ . If both endpoints lie in  $x(2E_{840}(\mathbb{Q}))$  then  $f(841+t)$  and  $f(841-t)$  are rational squares, so their product is one:  $w^2 = f(841+t)f(841-t)$ , which for  $a = 841$  simplifies to the genus-two curve

$$C : w^2 = (1-t^2)(841^2-t^2)(1681^2-t^2).$$

This is only a necessary condition, and that is enough: we close the superset. The even sextic makes  $C$  bielliptic. The quotient by  $(t, w) \mapsto (-t, w)$  has rank 3, as independently corroborated by the PARI/GP record shipped in the companion repository; this contextual calculation is not used in the proof and does not close the fibre. Thus the naive “compute this rank, then elliptic Chabauty” route was under-specified. The complementary quotient by  $(t, w) \mapsto (-t, -w)$  is decisive: for  $t \neq 0$  set  $X = 1/t^2$ ,  $Y = w/t^3$ , then with  $A = 841^2$ ,  $B = 1681^2$ ,  $L = AB$ ,  $u = LX$ ,  $v = LY$  the monic model is

$$E_- : v^2 = (u-A)(u-B)(u-L),$$

and  $\text{rank } E_-(\mathbb{Q}) = 0$  with  $E_-(\mathbb{Q})_{\text{tors}} \cong (\mathbb{Z}/2)^2$ , the nonzero torsion sitting at  $u \in \{A, B, L\}$ . The rank and torsion are computed in proof mode and independently replayed on the official Magma Calculator V2.29-8 (RankBounds = (0,0), proof flag true); the frozen transcript and machine-readable certificate ship in the companion repository, and Magma enters only as that hash-verified transcript. Every rational point of  $C$  with  $t \neq 0$  maps to  $E_-(\mathbb{Q})$ ; the three finite torsion points give  $t^2 = L/u \in \{1, A, B\}$ , the point at infinity of  $E_-$  lifts to the two points with  $t = 0$ , and the two infinite points of the even model are irrational since its leading coefficient is  $-1$ . Hence

$$t \in \{0, \pm 1, \pm 841, \pm 1681\}$$

for every affine rational point. Positive endpoint centres require  $841 \pm t > 840$ , i.e.  $-1 < t < 1$ , leaving only  $t = 0$ : the repeated three-entry boundary, not a distinct completion. The fibre contains no positive nonconstant three-transversal completion.  $\square$

### 5.3 A fixed Bremner shadow is constant

Bremner’s displayed degree-four solution [Bre01] decomposes into a rational constant part and a  $\sqrt{3}$ -anti-invariant part. After dividing by the constant centre, the resulting rational line is

$$J + tB, \quad B = \begin{pmatrix} 88 & -153 & 65 \\ -23 & 0 & 23 \\ -65 & 153 & -88 \end{pmatrix},$$

where  $J$  is the all-ones matrix. The next proposition concerns only this one line; it is not a global obstruction to magic squares of squares.

**Proposition 6** (Computer-assisted fixed Bremner shadow). *If every entry of  $J + tB$  is a square in  $\mathbb{Q}$ , then  $t = 0$ .*

*Proof.* The four conditions  $1 \pm 23t, 1 \pm 65t \in \mathbb{Q}^2$  imply

$$Y^2 = (1 - 23^2 t^2)(1 - 65^2 t^2).$$

For  $t \neq 0$ , putting  $x = 1/t^2$  and  $y = Y/t^3$  gives

$$E : y^2 = x(x - 529)(x - 4225).$$

Under the documented semantics of `ellrank`, the first two entries of the frozen PARI/GP output are unconditional lower and upper rank bounds; both are zero. The same record identifies the minimal model as Cremona curve 345345r4 [PARI, Cre]. For completeness, the torsion calculation is independently elementary. Reduction at 17 and 41 gives  $\#E(\mathbb{F}_{17}) = 16$  and  $\#E(\mathbb{F}_{41}) = 40$ , so rational torsion has order dividing 8. All three nonzero rational 2-torsion points fail the standard halving criterion: for the roots 0, 529, 4225, the two required root differences are never both rational squares. Hence  $E(\mathbb{Q})_{\text{tors}} = E[2](\mathbb{Q}) \cong (\mathbb{Z}/2)^2$ .

The finite  $x$ -coordinates are therefore 0, 529, 4225. Since  $x = 1/t^2$ , a nonzero parameter would have to be  $t = \pm 1/23$  or  $t = \pm 1/65$ . In each case one of the original four linear forms is 2, not a rational square. Thus no nonzero  $t$  is possible.  $\square$

## 5.4 The finite-class stop

*Proof of Theorem 5.* Well-definedness: if  $2Q' = P = 2Q''$  then  $Q' - Q'' \in E_d(\mathbb{Q})[2]$ , so the class of a half modulo  $2^n E_d(\mathbb{Q}) + E_d(\mathbb{Q})[2]$  does not depend on the half, and the doubling map exhibits the canonical isomorphism with  $2E_d(\mathbb{Q})/2^{n+1}E_d(\mathbb{Q})$  (kernel exactly  $2^n E_d(\mathbb{Q}) + E_d(\mathbb{Q})[2]$ ).

The countercertificate lives on  $E_{840}$  with the locked basis  $Q, R$  of Section 4. The two classical progressions have Kummer representatives  $\delta(Q) = (7, 10, 70)$  and  $\delta(R) = (14, 6, 21)$ , reproducing the rank-two structure. Now put  $H = Q + 2R$ : then  $\delta(H) = \delta(Q)$ , yet

$$x(2H) = \frac{490152611881}{53187849} \neq 841 = x(2Q),$$

and  $2H$  yields a genuinely distinct positive square progression (roots 667439/7293, 700109/7293, 731321/7293). In general, for every  $n \geq 1$  the points  $Q$  and  $Q + 2^n R$  share their  $\lambda_n$ -class while their doubled  $x$ -coordinates differ — equality would force  $2^{n+1}R = O$  or  $4Q + 2^{n+1}R = O$  against free independence. Finally compare the half triples  $(Q, Q, Q)$  and  $(Q, Q, Q + 2^n R)$ : identical  $\lambda_n$ -class triples, yet the first doubled-centre triple satisfies the interaction  $x_1 + x_2 = 2x_0$  and the second does not. Hence neither  $P \mapsto x(P)$  nor the interaction factors through the finite class triples, for any  $n$ .  $\square$

*Remark 6* (Scope of the stop). The certified true witness is deliberately the repeated triple  $(Q, Q, Q)$ , so the theorem rules out exactly the obstructions that retain only the three individual finite Kummer/2-Selmer classes on the ambient doubled-point space. It does not exclude representative-level methods — covering curves, heights, valuations, a Mordell–Weil sieve — nor a separate statement whose hypotheses first impose Parker-open positivity and pairwise distinctness. Sections 6 and 7 pursue precisely such finer invariants, along a different quantity.

## 6 The support law

This section proves Theorem 6. Throughout, a primitive positive Parker configuration is written with the three square progressions  $(r_i^2, s_i^2, t_i^2)$  of common difference  $d$ , and

$$u_i = t_i - r_i, \quad v_i = t_i + r_i, \quad z_i = u_i/v_i \in (0, 1), \quad u_i v_i = 2d.$$

### 6.1 The torus identity and nondegeneracy

Substituting  $u_i, v_i$  into the interaction  $s_1^2 + s_2^2 = 2s_0^2$  and using  $4s_i^2 = u_i^2 + v_i^2$  gives, after division by  $2d = u_i v_i$ , exactly

$$z_1 + z_1^{-1} + z_2 + z_2^{-1} = 2(z_0 + z_0^{-1}) : \tag{L.1}$$

multiplying by  $z_0/2$ , the five-term unit equation

$$\frac{z_0 z_1}{2} + \frac{z_0}{2z_1} + \frac{z_0 z_2}{2} + \frac{z_0}{2z_2} - z_0^2 = 1 \quad (\text{L.1}')$$

in the multiplicative group generated by the  $z_i$ .

*Nondegeneracy.* On the Parker open — all  $z_i \in (0, 1)$ , pairwise-distinct progressions — no nonempty proper subsum of the left side of (L.1') vanishes. Two exact facts drive every case. First, all  $z_i$  and  $z_i^{-1}$  lie in the squareclass of  $2d$ : indeed  $z_i \cdot 2d = u_i^2$  and  $2d/z_i = v_i^2$ , so any relation of the shape  $z_i = 2z_j$ ,  $z_i z_j = 2$ ,  $z_i = 2z_j^{-1}$  or  $2s_i^2 = u_0^2$  would make 2 a rational square. Second, the interaction itself. A vanishing subsum containing none of the four positive terms is empty; one containing only positive terms is positive. The remaining cases pair  $-z_0^2$  with one, two, or three positive terms. With *one*:  $z_0 z_1/2 = z_0^2$  gives  $z_1 = 2z_0$ , hence  $u_1^2 = 2u_0^2$  — excluded, and likewise for the other three singletons. With *three*: the subsum equals  $1 - (\text{omitted term})$  by (L.1') itself, so the omitted term equals 1, giving  $z_0 z_1 = 2$  or one of its three analogues — excluded the same way. With *two* same-index terms:  $z_1 + z_1^{-1} = 2z_0$  gives  $4s_1^2/(2d) = u_0^2/d$ , i.e.  $2s_1^2 = u_0^2$  — excluded. With *two* pure cross terms,  $z_1 + z_2 = 2z_0$ : combined with the interaction  $(u_1^2 + v_1^2) + (u_2^2 + v_2^2) = 2(u_0^2 + v_0^2)$  and  $v_i = 2d/u_i$ , the two relations force, for  $x = (u_0/u_1)^2$  and  $y = (u_0/u_2)^2$ , both  $x + y = 2$  and  $xy = 1$ , hence  $x = y = 1$  and  $z_1 = z_2 = z_0$  — against pairwise distinctness (and symmetrically for  $z_1^{-1} + z_2^{-1} = 2z_0$ ). With *two* mixed cross terms,  $z_1 + z_2^{-1} = 2z_0$ : the same elimination collapses to  $(u_0^2 u_2^2 - 4d^2)^2 = 0$ , i.e.  $u_0 = v_2$  and  $v_0 = u_2$ , so  $z_0 z_2 = 1$  — impossible with both in  $(0, 1)$ . Hence (L.1') is nondegenerate in the sense required by the subspace theorem.

## 6.2 The fixed-support count

Fix a finite prime set  $\mathcal{S} \ni 2$  and restrict to configurations with  $\text{supp}(2d) \subseteq \mathcal{S}$ . Modulo common rational-square scaling of the entries, the triple  $(z_0, z_1, z_2)$  determines the configuration, and each  $z_i$  lies in the group of  $\mathcal{S}$ -units of rank at most  $3|\mathcal{S}|$  relevant to (L.1). Theorem 1.1 of Evertse, Schlickewei and Schmidt [ESS02] bounds the nondegenerate solutions of a five-term unit equation over a group of rank  $r$  by  $\exp(30^{15}(r+1))$ ; applying it with  $r \leq 3|\mathcal{S}|$  gives at most

$$\exp(30^{15}(3|\mathcal{S}|+1))$$

root-primitive configurations. The bound counts; it does **not** enumerate, and no effective procedure follows from it.

## 6.3 Tropical and Legendre laws

For odd  $p$  the six valuations occurring in (L.1) are  $\pm\nu_i$  with  $\nu_i = v_p(z_i)$ ; in a vanishing non-Archimedean sum the least valuation is attained at least twice, which proves that the largest  $|\nu_i|$  occurs at least twice. At  $p = 2$  the same argument applies to  $\min\{\nu_1, -\nu_1, \nu_2, -\nu_2, 1 + \nu_0, 1 - \nu_0\}$ . Writing  $e = v_p(2d)$  and  $a_i = v_p(u_i)$ , the relation  $u_i v_i = 2d$  gives  $\nu_i = 2a_i - e$  with  $0 \leq a_i \leq e$ ; call an index an *endpoint* if  $a_i \in \{0, e\}$  and *interior* if  $0 < a_i < e$ . For odd  $p$ , an interior index puts  $p$  in every root of its progression:  $p$  divides both  $u_i = t_i - r_i$  and  $v_i = t_i + r_i$ , hence  $t_i$  and  $r_i$ , and then  $4s_i^2 = u_i^2 + v_i^2$  forces  $p \mid s_i$ . Root primitivity therefore forbids three interior indices, so at least one index is an endpoint; and an endpoint attains the maximal value  $|\nu_i| = e$ , so the tropical tie forces a second endpoint: **at least two indices are endpoints at every odd support prime**. If exactly two indices are endpoints, comparing the unit squareclasses of the two least-valuation terms forces  $-1$  to be a quadratic residue for the pair  $\{1, 2\}$  and 2 for a pair containing 0; hence

the exact endpoint pairs obey

| $p \bmod 8$ | allowed exact endpoint pairs          |
|-------------|---------------------------------------|
| 1           | $\{0, 1\}, \{0, 2\}, \{1, 2\}$        |
| 3           | none; all three indices are endpoints |
| 5           | $\{1, 2\}$ only                       |
| 7           | $\{0, 1\}, \{0, 2\}$ only             |

These are necessary local laws, not sufficient conditions.

#### 6.4 Exclusion of support $\{2, 3\}$

By Lemma 1 of [PTZ15] every entry of a primitive configuration is  $1 \bmod 24$ , so all roots are coprime to 6 and  $24 \mid d$ . Assume  $\text{supp}(2d) \subseteq \{2, 3\}$  and put  $e_p = v_p(2d)$ . At 2, both  $u_i, v_i$  are even and exactly one has valuation 1, so  $v_2(z_i) = \pm(e_2 - 2)$ ; at 3, exactly one of them is divisible by 3, so  $v_3(z_i) = \pm e_3$ ; there are no other primes. Hence  $z_i \in \{2^{\pm(e_2-2)}3^{\pm e_3}\}$ , and since  $z \mapsto z^{-1}$  pairs the four sign choices,  $f(z) = z + z^{-1}$  takes at most two values on the  $z_i$ . Three elements of a two-element set can satisfy  $f(z_1) + f(z_2) = 2f(z_0)$  only if all three values agree, and  $f$  is strictly decreasing on  $(0, 1)$ , so  $z_0 = z_1 = z_2$ ; with the same  $K = 2d$  the formulas  $u_i = \sqrt{K}z_i, v_i = \sqrt{K}/z_i$  then force the three progressions to coincide, contradicting pairwise distinctness. This completes the proof of Theorem 6.

*Remark 7* (Scope). Everything here is a necessary-condition layer on the common difference. The subspace-theorem count is imported [ESS02]; the congruence input is [PTZ15]; the tropical mechanism is elementary. No effectivity, no global support-size bound and no all-support emptiness follows, and none is claimed from this counting argument. The separate two-term  $S$ -unit route in Section 8 does give a finite enumeration for the specific complete support catalogues used there.

### 7 Three-prime exclusion

*Proof of Theorem 7.* Divide the roots of a positive square progression by their gcd  $g$ . The primitive roots  $r', s', t'$  are all odd (reduce modulo 4), and  $U = (t' - r')/2, V = (t' + r')/2, s'$  form a primitive right triangle:  $U^2 + V^2 = s'^2, \gcd(U, V) = 1$ . Its area  $A = UV/2$  satisfies  $d/4 = Ag^2$ , so its prime support has at most three elements.

Aebi's theorem and closing remark [Aeb26] classify all such primitive triangles, up to exchanging the legs. Their areas and squarefree parts are, respectively,

| $A$            | 6 | 30 | 60 | 180 | 84 | 504 | 1224 |
|----------------|---|----|----|-----|----|-----|------|
| $\text{sf}(A)$ | 6 | 30 | 15 | 5   | 21 | 14  | 34   |

and each area has exactly one primitive triangle. The squarefree parts are distinct. If two positive progressions had the same  $d$ , the ratio of their primitive areas would be a rational square, forcing the same area and the same  $g$ . The triangle then recovers the unique roots  $g|U - V|, gs', g(U + V)$ , so the progressions coincide. This proves the statement and its array consequences. The bound on the number of primes cannot be raised to four for this uniqueness statement: at  $d = 840$  the two triples  $(1, 29, 41)$  and  $(23, 37, 47)$  are distinct examples.  $\square$

#### 7.1 The elementary endpoint route

For comparison, we retain the local endpoint argument and valuation restrictions. In the rest of this section the configuration is primitive and satisfies the full-magic interaction.

Throughout,  $\text{supp}(2d) = \{2, 3, p\}$  with  $p \neq 3$  prime, and  $d/2 = XYZ$  with  $X = 2^\alpha$ ,  $Y = 3^\beta$ ,  $Z = p^\gamma$ . The two technical stages are stated first; the headline Theorem 7 subsumes both by a different global argument, and its proof is independent of theirs.

**Proposition 7** (All-endpoint case,  $p \equiv 3 \pmod{8}$ ). *There is no primitive positive Parker configuration with  $\text{supp}(2d) = \{2, 3, p\}$  and  $p \equiv 3 \pmod{8}$ .*

*Proof.* By the endpoint table of Section 6,  $p \equiv 3 \pmod{8}$  forces all three indices to be endpoints at  $p$ , at 3 (same residue argument), and — by the opposite-parity mechanism at 2 — at the 2-power as well. Each progression's equal-area Pythagorean triangle is then primitive, and its legs are a unitary partition of the three whole blocks  $X, Y, Z$ . Apart from the impossible split  $\{1, XYZ\}$ , only  $\{X, YZ\}$ ,  $\{Y, XZ\}$ ,  $\{Z, XY\}$  remain, and three pairwise-distinct progressions must use all three. Factoring the hypotenuse equations of the last two gives

$$Y^2 - 1 = 2XZ, \quad Z^2 - 1 = 2XY,$$

whose difference factors as  $(Y - Z)(Y + Z + 2X) = 0$ . Positivity forces  $Y = Z$ , after which  $Y(Y - 2X) = 1$ , impossible for positive prime-power blocks.  $\square$

**Proposition 8** (One-interior law,  $p \equiv 5, 7 \pmod{8}$ ). *Let  $\text{supp}(2d) = \{2, 3, p\}$  with  $p \equiv 5$  or  $7 \pmod{8}$ . The all-endpoint branch is impossible by the three-block argument, so  $\gamma = 1$  is excluded, and for  $\gamma \geq 2$  exactly one index is interior at  $p$ . If  $p \equiv 5 \pmod{8}$  the centre progression is the interior one and exactly one of*

$$v_p(X^2 + Y^2) = 2h, \quad v_p(1 + Y^2) = 2h, \quad v_p(1 + X^2) = 2h$$

*holds, with  $h = v_p(s_0) \geq 1$ ; if  $p \equiv 7 \pmod{8}$  one outer progression is interior and exactly one of the six ordered differences*

$$v_p(2X^2 - Y^2), v_p(2Y^2 - X^2), v_p(2 - Y^2), v_p(2 - X^2), v_p(2Y^2 - 1), v_p(2X^2 - 1)$$

*equals  $2h$ , with  $h = \min(a, \gamma - a)$ . These are necessary branch laws, not contradictions; their exact certificate covers all nine identities.*

The nine residual branches of Proposition 8 need not be analyzed one by one: the headline theorem closes every case, including  $p \equiv 1 \pmod{8}$ , at once.

*An elementary proof for primitive exact support  $\{2, 3, p\}$ .* By the endpoint table, every index is an endpoint at 3 and at least two indices are endpoints at  $p$ . Those two indices give two *distinct primitive* Pythagorean triangles of the same area  $d/4$ , whose area has exactly three distinct prime factors. Writing  $X = 2^\alpha = 2x$ , the three endpoint types satisfy

$$YZ = x^2 - 1, \quad Y^2 - 1 = 4xZ, \quad Z^2 - 1 = 4xY.$$

Combining the first with each of the others reduces both mixed pairs to one equation

$$W(W^2 - 1) = 4x(x^2 - 1),$$

where  $W$  is the remaining odd block. The 2-adic valuation of the right side forces  $W \equiv \pm 1 \pmod{2x}$ , while monotonicity of  $W(W^2 - 1)$  gives  $x < W < 2x$ ; the only residue in that window is  $W = 2x - 1$ , which forces  $x = 2$ ,  $W = 3$ , and then the third block equals 1 — not a prime power in the support. The remaining pair of endpoint types forces an equality of powers of two distinct primes, equally impossible. Hence no primitive positive Parker configuration has  $\text{supp}(2d) = \{2, 3, p\}$  for any prime  $p \neq 3$ .  $\square$

*Remark 8* (Attribution and scope). The headline theorem is the squareclass consequence of Aebi's complete triangle classification proved at the start of this section. The endpoint argument gives another proof of its primitive full-magic consequence, without a priority claim. The uniqueness assertion fails already at four primes, as the difference-840 example shows. Section 8 nevertheless closes particular four-prime supports by a separate complete catalogue; arbitrary four-prime supports and global support-size control remain open.

## 8 Complete catalogues for small prime supports

The counting theorem of Section 6 is not itself an enumeration algorithm. A different, effective route is available for a *fixed* support: solve a two-term  $S$ -unit equation, reconstruct primitive right triangles, and test their normalized centres. We execute this route for  $S_B = \{p \text{ prime} : p \leq B\}$  with  $B = 7, 13, 19$ . Completeness depends explicitly on an imported computer-assisted theorem; the finite calculations below do not re-prove that theorem.

### 8.1 The imported total and its normalization

Let  $\mathcal{A}(S)$  be the set of integer triples

$$0 < a \leq b < c, \quad a + b = c, \quad \gcd(a, b) = 1, \quad \text{supp}(abc) \subseteq S.$$

Theorem A of von Känel and Matschke [vKM16], printed p. 7, counts solutions of  $x + y = 1$  in rational  $S$ -units modulo the six usual symmetries. Each orbit has exactly one representative  $0 < x \leq 1/2$ ; writing it as  $x = a/c$ ,  $1 - x = b/c$  gives exactly one element of  $\mathcal{A}(S)$ . The exceptional orbit represented by  $(1, 1, 2)$  is counted once. Thus their entries for the first four, six and eight primes give

$$|\mathcal{A}(S_7)| = 63, \quad |\mathcal{A}(S_{13})| = 545, \quad |\mathcal{A}(S_{19})| = 3649.$$

The first two totals were already obtained by de Weger, as credited in [vKM16]. We import only these three totals, not a conjecture comparing different supports of the same size.

The supplied producer finds 3649 distinct valid triples with  $c \leq 10^{10}$ . The separate verifier checks integrality, positivity, ordering, coprimality, addition and prime support for every row, and checks the imported total. A subset of a finite set with the same cardinality is the entire set: this is the completeness argument, with *no remaining height restriction*. The production bound alone would prove no such claim. The witnesses are regenerated locally, not copied from the authors' data files; their source theorem remains a mathematical dependency.

### 8.2 From additive triples to square progressions

**Lemma 5** (Exact triangle reconstruction). *Primitive positive integer right triangles, up to swapping the legs, whose area is supported on  $S$  correspond bijectively to pairs  $m > k > 0$  with  $\gcd(m, k) = 1$ ,  $m + k$  odd, and*

$$(\min(k, m - k), \max(k, m - k), m) \in \mathcal{A}(S), \quad (k, m, m + k) \in \mathcal{A}(S).$$

*Their legs, hypotenuse and area are*

$$\{a, b\} = \{2mk, m^2 - k^2\}, \quad c = m^2 + k^2, \quad A = mk(m - k)(m + k).$$

*Proof.* Euclid's parametrization is unique with these coprimality and parity conditions. Its area is supported on  $S$  exactly when all four positive factors  $m, k, m - k, m + k$  are supported on  $S$ . The two displayed additive triples express precisely these conditions; their coprimality follows from  $\gcd(m, k) = 1$ . Conversely, membership and odd  $m + k$  give Euclid's conditions and the required area. This proves both directions, not just a procedure for generating examples.  $\square$

Write  $A = nh^2$  with  $n$  squarefree. The corresponding normalized square progression has positive rational roots

$$\left( \frac{|a-b|}{h}, \frac{c}{h}, \frac{a+b}{h} \right), \quad d = 4n, \quad \sigma = \frac{c^2}{h^2} > 4n.$$

Conversely, divide any positive integer-root square progression by the gcd  $g$  of its roots. Write the divided roots as  $r', s', t'$ ; they are odd, and  $((t' - r')/2, (t' + r')/2, s')$  is a primitive right triangle of area  $d/(4g^2)$ . Thus  $d/4 = n(hg)^2$ . Progressions of the same difference necessarily have the same  $n$  and common scale  $\lambda = hg$ , so their actual centres are  $\lambda^2\sigma$ . Consequently the full-magic interaction is exactly

$$\sigma_1 + \sigma_2 = 2\sigma_0.$$

All denominators of the normalized roots divide  $h$  and are supported on  $S$ . Clearing them and dividing by the common root gcd therefore gives every root-primitive integral array without introducing a prime outside  $S$  into its difference.

### 8.3 The complete finite tests

*Proof of Theorem 8.* The complete additive lists and Lemma 5 give the following exact counts. A midpoint test checks one unordered pair of distinct centres within the same squarefree area class and asks whether their average is another centre in that class.

| $B$ | $ \mathcal{A}(S_B) $ | triangles | classes | pairs | hits | $(9, 7)/D_4$ classes |
|-----|----------------------|-----------|---------|-------|------|----------------------|
| 7   | 63                   | 12        | 9       | 4     | 0    | 3                    |
| 13  | 545                  | 62        | 39      | 36    | 0    | 45                   |
| 19  | 3649                 | 232       | 133     | 180   | 0    | 384                  |

Every number in this table is recomputed with integer and rational arithmetic by the companion's exact checker:

`scripts/verify_smooth_support.py.`

In particular, the normalized centres within each class are distinct and no midpoint test succeeds. By the preceding equivalence, no fully magic square in the stated domain exists. This establishes part (i).

For part (ii), group the triangles by  $n$  and select three distinct triangles in one group. The three normalized progressions give nine candidate entries. Reject nonpositive or repeated entries; for these catalogues none of the 1, 15, 128 unordered triples is rejected. Choose one of the three centres as the central cell  $E$ , use the other two as  $A, P$  in the normal form, and set  $D = 4n$ . The absence of midpoint hits ensures exactly seven equal line sums. Clear root denominators, divide by the root gcd, and take the lexicographically least positive-root array in its eight-element  $D_4$  orbit.

These operations are exhaustive by the normal form, which also holds when all eight lines agree. For an array with exactly one failed line, orienting that diagonal as the antidiagonal leaves

precisely the choices of interchanging  $A, P$  and changing the sign of  $D$ . Those choices are already identified by  $D_4$ ; the central cell is fixed by every grid symmetry. Therefore the three choices of  $E$  are inequivalent. The unique primitive normalization of a positive rational-root array also removes all common rational-square scaling. The verifier reconstructs and compares all 3, 45, 384 representatives, establishing the classification.

Finally, for each representative compute the seven-line sum  $M$ , the failed sum  $F$ , and

$$\eta = \frac{|M - F|}{|D|} = \frac{|A + P - 2E|}{|D|}.$$

This quotient is invariant under  $D_4$  and common square scaling. Exact comparison over the complete lists gives minima 1727/1120 for  $S_7$  and 347984603/9837828000 for both  $S_{13}$  and  $S_{19}$ , each attained by one class. For the latter, a primitive root representative (the *entries of the array are its squares*) is

$$\begin{pmatrix} 87942 & 140595 & 157690 \\ 186290 & 132558 & 9555 \\ 99645 & 122590 & 165558 \end{pmatrix}.$$

It has  $|D| = 9837828000$ ,  $M = 52366885489$ ,  $F = 52714870092$ , and  $M - F = -347984603$ . These identities and uniqueness are checked from the full atlas, proving part (iii). No floating-point comparison is used.  $\square$

For  $S_7$  the unique unordered triple, after common root-primitive normalization, has roots (2, 58, 82), (46, 74, 94) and (97, 113, 127) and common difference 3360. Its three choices of the central progression account for the three classes. The atlas is for (9, 7) arrays; it is separate from the bounded (8, 7) census of Appendix A. Also,  $\eta$  is normalized by the progression difference, not by the magic sum: it is not a percentage error or a comparison with unrestricted near-miss records.

## 8.4 Consequences and limits

Every hypothetical fully magic square of distinct positive integer squares must have a prime at least 23 in each of its two transversal differences. Indeed, write its normal form as  $M(E - t, E, E + t; d)$ , where  $M$  denotes the matrix in Theorem 1. A clockwise quarter-turn gives  $M(E + d, E, E - d; t)$ . Both differences are nonzero, and applying part (i) in each orientation proves the assertion. The two large primes need not be different. These are primes of the *differences*, not of the entries.

In particular, the fifteen supports consisting of 2, 3 and two distinct primes chosen from  $S_{19} \setminus \{2, 3\}$  are excluded. This does not close arbitrary four-prime supports, bound the number of support primes, or settle the general magic-square-of-squares problem. The general fixed-squareclass finiteness result of Section 5 remains non-effective; the present enumeration is obtained through the separate fixed-support route and has been executed only for the declared catalogues.

## 9 The four-prime frontier

Section 8 closes all supports contained in  $S_{19}$ . Here the extra primes are arbitrary; the remaining branches are not resolved by that finite-support result.

Let  $\text{supp}(2d) = \{2, 3, p, q\}$  with  $p, q$  distinct primes different from 2, 3, and put  $U_i = (t_i - r_i)/2$ ,  $V_i = (t_i + r_i)/2$ ,  $N = d/2$ , so  $U_i V_i = N$  and  $U_i^2 + V_i^2 = s_i^2$ .

**Proposition 9** (Endpoint incidence and edge factorization). *Each extra prime labels either no interior progression or one residue-permitted interior vertex, so at least one progression is endpoint at both  $p$  and  $q$  and its Pythagorean triple is primitive; the number of primitive progressions is  $3 - \#(\{I_p, I_q\} \setminus \{\emptyset\})$ . For every pair  $i, j$ , setting*

$$G = \gcd(U_i, U_j), \quad R = U_i/G, \quad S = U_j/G, \quad B = N/(GRS)$$

*gives  $U_i = GR$ ,  $V_i = BS$ ,  $U_j = GS$ ,  $V_j = BR$  and the edge identity*

$$s_i^2 - s_j^2 = (G^2 - B^2)(R^2 - S^2).$$

*An interior-prime label splits its prime-power block on the two incident edges and leaves it whole on the opposite edge; the Parker interaction becomes the equality  $F_{10} = F_{02} = D \neq 0$  of the two canonical products of four difference factors. Two exact fixtures (area 210; the mixed area-30600 dropout) certify that neither two primitive progressions nor one primitive plus one interior progression is by itself contradictory — which is precisely why no exclusion is claimed at this level.*

**Proposition 10** (Balanced-dropout reduction). *If an extra prime is balanced at an interior vertex (its exponent split equally between the two legs), every other interior label is absent or co-located at the same vertex; every balanced branch has  $d/4 = Ag_i^2$  with*

$$A \in \{6, 30, 60, 180, 84, 504, 1224\},$$

*and primitive core  $(3, 4, 5)$  or one of the six cores classified by Aebi [Aeb26]. If both extra primes are balanced at the central vertex, then  $A = 6$  and, after scaling, a candidate forces a rational point on*

$$w^2 = (1 - t^2)(625 - t^2)(2401 - t^2),$$

*whose complementary bielliptic quotient (the method of Flynn and Wetherell [FW99]) is  $y^2 = x(x - 111)(x - 93750)$ , of rank zero with torsion  $(\mathbb{Z}/2)^2$  — proof-mode computation replayed on the official Magma V2.29-8, entering as a hash-verified frozen transcript. The rational lifts give  $t \in \{0, \pm 1, \pm 25, \pm 49\}$ , positivity requires  $|t| < 1$ , and  $t = 0$  repeats the progression: the central double-balanced branch is empty.*

**Proposition 11** (Outer core-area-6 exclusion and the addition curves). *If both extra primes are balanced at an outer vertex, the primitive core is  $(3, 4, 5)$ ; with core entries 1, 25, 49 and central and opposite outer entries  $25 + t$ ,  $25 + 2t$ , a candidate forces a rational point on*

$$W^2 = (t + 1)(t + 25)(t + 49)(2t + 1)(2t + 25)(2t + 49),$$

*whose complete set of rational  $t$ -coordinates is*

$$\{-49, -25, -49/2, -25/2, -1, -1/2, 0\}$$

*(certified-complete point list, frozen transcript). Positivity of both new progressions requires  $t > -1/2$ , so only  $t = 0$  survives and repeats the core: the outer double-balanced core-area-6 branch is empty on the positive distinct locus. For the residual central cores, normalizing the congruent-number curve as  $E_n: y^2 = x^3 - n^2x$  with fixed central  $x$ -coordinate  $a$ , a secant slope  $z$  between  $P, Q$  with  $x(P) + x(Q) = 2a$  lies on*

$$H_{n,a}: Y^2 = (z^2 - (2a - n))(z^2 - 2a)(z^2 - (2a + n));$$

*for the six Aebi central cores (areas 30, 60, 180, 84, 504, 1224) the Jacobian ranks are 2, 2, 3, 3, 4, 5, so ordinary genus-two Chabauty does not apply. In the first two cases, two-cover descent leaves a singleton fake two-Selmer set, identifying a unique covering class for elliptic or quadratic Chabauty in the sense of Bruin [Bru03] — a target, not a result: the rational points remain undetermined.*

## The exact open boundary

**No arbitrary four-prime exclusion is claimed.** What remains open at this frontier, stated exactly: the six Aebi central genus-two fibres (ranks 2, 2, 3, 3, 4, 5 above), the outer fixed cores other than the area-6 one, and the persistent unbalanced four-prime primitive cores. The continuation of the residual arithmetic growing out of the area-30/60 covering classes lives in the companion repository’s dossiers with its own open gates, and no statement about it is imported here.

## 10 Conclusions and the computational program

The theorems exclude every difference with at most three prime factors, close the full-magic problem for differences supported on primes at most 19, classify the associated (9, 7) arrays, and close one interaction fibre. They leave the arbitrary-support frontier open. This closing section summarizes the program that organizes what remains, and states where each open object lives. Everything here is a summary; the companion repository’s self-contained dossiers and exhausted-routes catalogue give the precise public boundaries.

### 10.1 The difference-side taxonomy

For each progression write  $U_i = (t_i - r_i)/2$ ,  $V_i = (t_i + r_i)/2$ , so  $U_i V_i = d/2$  and  $U_i^2 + V_i^2 = s_i^2$ . An odd support prime is interior at most at one vertex, and its allowed interior vertices depend only on  $p \bmod 8$ : for  $p \equiv 1$  all of  $\{0, 1, 2\}$ ; for  $p \equiv 3$  none; for  $p \equiv 5$  only the centre  $\{0\}$ ; for  $p \equiv 7$  only the outer vertices  $\{1, 2\}$ . The number of primitive progressions is  $3 - \#\{\text{distinct occupied interior vertices}\} \geq \max(0, 5 - \omega(2d))$ . The eight labelled incidence strata have exact arithmetic minima for the common difference (24, 600, 1,176, 29,400, 339,864, 8,496,600 for the deepest stratum), and the zero-primitive stratum passes its combinatorial gate exactly under the Hall conditions

$$n_1 + n_5 \geq 1, \quad n_1 + n_7 \geq 2, \quad n_1 + n_5 + n_7 \geq 3$$

on the counts of extra primes by residue class. **These are necessary minima and matching conditions, not existence results**, and no Diophantine statement follows from the taxonomy alone. The entry-side counterpart of this classification — which primes divide which *entries* — is the lineage of Rabern, Woll, Labruna and Weisenberg [Rab03, Wol18, Lab18, Wei23] discussed in Section 1. On the centre side, the catalogue of Gaussian representations of the square centre parallels the signature-based organization proposed by Weisenberg [Wei23], §3.

### 10.2 The compiler and the outer reduction

A fail-closed compiler accepts the complete factorization of  $2d$  with the three valuation rows, raises a typed violation of any encoded necessary law, routes exact supports  $\{2, 3\}$  and  $\{2, 3, p\}$  to the exclusions of Sections 6–7, and otherwise emits one canonical branch key with its successor problem — an interface closure, deliberately not a Diophantine one. For the two-primitive-AP outer branch, the canonical edge factors are pairwise coprime with exactly one divisible by four; the two normalized odd norms share a unique squarefree kernel; every odd prime common to them is 1 mod 8; and the branch admits an explicit double-cover presentation — the entry point of the surface geometry of Section 4.

### 10.3 Where the open objects live

The companion repository carries three dossiers, each declaring its terminal state exactly:

- **Residual cover:** the descent chain growing out of the area-30/60 covering classes of Section 9, developed through explicit higher descents to a single residual class  $q_R$  over  $\mathbb{Q}(\sqrt{218})$ ; open at exactly the decision  $q_R(K)$  — a point gives rank two, emptiness gives rank one with a nontrivial divisible Tate–Shafarevich class.
- **The lift tower** ( $n \geq 3$ ): the reconstruction covers of the transversal frame; the  $n = 3$  branch is closed completely in the dossier, the  $n = 5$  branch is developed to two genus-two covers over  $\mathbb{Q}(\sqrt{2})$  whose complete fake two-Selmer set has four classes, none eliminated, with a saturated rank-two pair-field lattice and conditional elliptic-Chabauty gates.
- **The degree-90 toolchain:** the class of a marked half ideal  $[A_R]$  in a degree-90  $S$ -class group, open, with the field itself now computable, the splitting  $d_{90} = \dim \text{Cl}_S(F_{30})[2] + \dim \ker(N)[2]$  proved, and the short-vector route measured to exhaustion.

An exhausted-routes catalogue records every closed route with its exact boundary, so that later work need not re-run a dead end. The deepest remaining problems — deciding every one-primitive branch, closing the zero-primitive high-support stratum by an effective coloured  $S$ -unit theorem, and the final exhaustive join — remain open as uniform problems over varying supports. For a fixed support the two-term  $S$ -unit route of Section 8 provides a finite reduction; this does not settle the all-support join.

## A Census and provenance

### A.1 The exact minimum census

The census definition is frozen: nine distinct positive integer entries; exactly eight positive squares; exactly seven of the eight ordinary lines with one common sum; quotient by the geometric  $D_4$  action; height equal to the largest positive square root among the eight squares.

**Proposition 12** (Exact minimum). *Under this definition there are 0 classes of height  $\leq 46$  and exactly 9 classes of height  $\leq 47$ , of which exactly two have a nonsquare centre.*

*Certificate argument.* Two independently implemented exact enumerators in the companion repository agree: `census.reference.py` scans integer progression centres and common differences, while the census family of `verify.py` builds full progressions from root triples and partial ones from root pairs. At height 46 there are twelve full three-square progressions, all with distinct common differences; two full progressions of one difference are necessary, so the census is empty. At height 47 the only difference supporting two full progressions is 840:

$$(1, 841, 1681) = (1^2, 29^2, 41^2), \quad (529, 1369, 2209) = (23^2, 37^2, 47^2),$$

There are five raw partial progressions satisfying the square-height bound. Two repeat entries when combined with the full progressions and therefore fail nine-way distinctness:

$$(841, 1681, 2521), \quad (1369, 2209, 3049).$$

The only compatible partial progressions are

$$(121, 961, 1801), \quad (169, 1009, 1849), \\ (256, 1096, 1936).$$

For each partial progression the six role assignments collapse to three  $D_4$  classes, giving  $3 \times 3 = 9$ ; the last two partials carry their nonsquare in the middle, and assigning either to the centre role gives one centre-nonsquare class each. The census is replayed from scratch by the companion verifier (`verify.py`, census family), and all nine canonical representatives are frozen in its expected certificate.  $\square$

The three shifts  $t = 9, 96, 144$  of the  $n = 1$  carrier antidiagonal  $(265, 1105, 1945)$  produce exactly the three partial components of the minimum layer — stated carefully: only three designated orientations are literal antidiagonal retractions of the fixed source orientation; the other six classes are role reassignments. **The public census is frozen at the height-47 bound above**; no figure at any other bound is claimed here.

## A.2 Provenance of the centre-true eight-square arrays

A related catalogued family must not be confused with the  $(8, 7)$  metric: the *centre-true* arrays with a nonsquare centre  $g$ , eight distinct square entries around it, magic sum  $3g$ , and **six** magic lines (three rows, middle column, both diagonals; the outer columns fail). Seven such arrays with centres

$$157441, 411625, 786361, 1199809, 2033641, 12138289, 19953649$$

circulated in an earlier phase of this project as if new. A source audit corrected the record: Brown’s note on automedian triangles and magic squares [Bro] already lists, for each of these seven centres, the six roots of the top and bottom rows. What this project adds is only the square middle-pair completion and the six-line arrangement; whether that completion is itself absent from the literature has not been source-locked, and no novelty is claimed. For example, at  $g = 157441$ :

$$\begin{pmatrix} 421^2 & 49^2 & 541^2 \\ 481^2 & g & 289^2 \\ 149^2 & 559^2 & 371^2 \end{pmatrix},$$

whose six stated lines each sum to  $3g = 472323$ . Brown also states that a Parker centre root needs more than two distinct primes  $1 \bmod 4$ ; that statement is non-refereed and its proof scope is unaudited here, but it blocks any “first centre-prime obstruction” wording, and none appears in this paper.

## B The least-shift interval analysis, and certificates

### B.1 The complete case analysis for Proposition 3

Setting and notation as in Section 3:  $n \geq 2$ ,  $(y, z) \mapsto (y + 2z, y + 3z)$  from  $(11, 15)$ , induction inequalities

$$z > y, \quad 2y > z, \quad 5y > 3z, \quad 5z > 6y, \quad 8y > 5z, \quad z \geq 15,$$

$t_0 = 6z^2$ ,  $h = t_0 - t$  with  $0 < h < 6z^2$ , and at  $t_0$  the three entries are  $C - t_0 = Y^2$  ( $Y = yz + 1$ ),  $E - t_0 = H = 4z^4 - z^2 + 1$ ,  $G - t_0 = X^2$  ( $X = xz - 1$ ).

*The C entry.* If  $C - t = (Y + a)^2$  then  $h = a(2Y + a)$ . If  $a \geq 5$  then  $h \geq 5(2Y + 5) = 10yz + 35 > 6z^2$  because  $5y > 3z$ ; hence only  $a \in \{1, 2, 3, 4\}$ .

*The G entry.* If  $G - t = (X + b)^2$  with  $b \geq 2$  then  $h \geq 4X + 4 = 4yz + 8z^2 > 6z^2$  (here  $X = xz - 1$  and  $x = y + 2z$  along the recurrence); thus  $b = 1$  and necessarily  $h = 2X + 1 = 2yz + 4z^2 - 1$ .

*The E entry.* The exact neighbouring squares are

$$(2z^2 - 1)^2 < H < (2z^2)^2 < (2z^2 + 1)^2 < H + 6z^2 < (2z^2 + 2)^2,$$

so  $E - t$  can be square only for  $h = z^2 - 1$  or  $h = 5z^2$ .

*Excluding CE.* Equating  $a(2Y + a)$  with the two possible  $E$  increments gives, for  $a = 1, 2, 3, 4$  respectively:

| $a$ | equality with $z^2 - 1$ | equality with $5z^2$ |
|-----|-------------------------|----------------------|
| 1   | $z(z - 2y) = 4$         | $z(5z - 2y) = 3$     |
| 2   | $z(z - 4y) = 9$         | $z(5z - 4y) = 8$     |
| 3   | $z(z - 6y) = 16$        | $z(5z - 6y) = 15$    |
| 4   | $z(z - 8y) = 25$        | $z(5z - 8y) = 24$    |

The first column is negative in every row because  $2y > z$ . In the second column, rows 1 and 2 equate positive multiples of  $z \geq 15$  with 3 or 8; row 3 has  $5z - 6y > 0$ , and for  $z > 15$  the product exceeds 15 while at  $(y, z) = (11, 15)$  it equals 135; row 4 is negative because  $8y > 5z$ . Branch  $CE$  is impossible.

*Excluding CG.* Equating the four  $C$  increments with  $2yz + 4z^2 - 1$  gives respectively

$$4z^2 = 4, \quad 2z(2z - y) = 9, \quad z(z - y) = 4, \quad 2z(2z - 3y) = 25.$$

The first and third contradict  $z \geq 15$ ; the second and fourth equate an even integer with an odd one. Branch  $CG$  is impossible.

*Excluding EG.* The  $G$  increment exceeds  $z^2 - 1$ , their difference being  $2yz + 3z^2 > 0$ ; equality with  $5z^2$  would give  $z(2y - z) = 1$ , contradicting  $z \geq 15$  and  $2y > z$ . Branch  $EG$  is impossible.

Every pair is excluded, so no positive pair-closing shift is smaller than  $6z^2$ ; the shift  $6z^2$  itself closes  $C$  and  $G$ , and Theorem 2(i) shows it is nondegenerate. The proof is uniform in  $n$ , uses no factorization of a particular  $\Delta$  and no scan over  $t$ ; the packaged factor-pair census for  $1 \leq n \leq 6$  is corroboration only. The statement deliberately starts at  $n = 2$ : at  $n = 1$  the shift 9 is smaller than the uniform  $6z^2 = 96$ .

## B.2 Certificate discipline

The companion repository distinguishes two verification layers. Its single verifier entry point, implemented in Python, re-derives the elementary and symbolic witnesses (the integral line-sum Smith data; the recurrence identities and least shifts; the bridge divisor check and the (79, 65, 89) completion; the elliptic-orbit witnesses with denominator clearing; the saturation and monodromy data; the fibre algebra and pulled-back  $t$ -lists; the fixed Bremner-shadow quotient, finite-field counts and torsion-halving test; the half-class countercertificate; the torus, endpoint-table and three-block identities; the four-prime incidence and fixed-fibre identities) using exact integer and symbolic arithmetic. A deterministic high-precision decimal quadrature corroborates only the printed decimal of the exact Haar-measure density formula; it is not a logical input to the density proof. Computations performed using external computer algebra (the rank, torsion, complete point-list and Selmer gates attributed in the text to the official Magma Calculator V2.29-8, plus two PARI/GP quotient records) enter as *hash-pinned inputs and normalized output records*. These

frozen transcripts and records are authenticated by the verifier, which checks their hashes and parsed assertions; it does not independently prove those external calculations and never executes either system. The fixed Bremner-shadow rank is supplied by a hash-pinned PARI/GP input/output pair; the verifier independently checks every elementary step surrounding that rank result. The exact file-level map is generated with the verifier and covered by the repository manifest.

## C Reproducibility

### C.1 Engines

All theorem-level arithmetic in this paper is exact: integer and symbolic computation in Python with implementation-separated replays where a result is load-bearing, and elliptic-curve rank, torsion, and rational-point computations in proof mode. Results attributed to the official Magma Calculator V2.29-8 (the rank-zero quotients of Sections 5 and 9, the certified-complete point lists, the descent transcripts) enter the record exclusively as **hash-verified frozen transcripts**: the verifier checks digests and parsed assertions and never executes that software. One accompanying PARI/GP record corroborates only the non-load-bearing rank-three quotient of Section 5; a second pinned record supplies the unconditional rank-zero interval for Proposition 6, while the verifier independently checks its quotient map, finite-field counts, torsion and terminal parameter elimination. No floating-point quantity feeds any conclusion. The checked-in manuscript was built deterministically with MiKTeX-pdfTeX 4.23 (MiKTeX 25.12), using the three-pass `pdflatex` route at the fixed `SOURCE_DATE_EPOCH`; the source also supports the Tectonic route documented in the companion repository. It is distributed as a public preprint companion artifact.

### C.2 The smooth-support certificates

The complete total 3649 in Theorem 8 is imported from von Känel–Matschke [vKM16]; the corresponding witnesses and all downstream triangles, midpoint tests and  $(9, 7)$  arrays are rebuilt locally. The producer and verifier do not import each other’s code: one enumerates  $k + (m - k) = m$  and tests  $m + k$ , while the other starts with  $k + m = m + k$  and joins the other additive record. Their  $D_4$  canonicalizations are also separate. These checks validate the finite deduction, not the source theorem’s original height bounds or sieve.

### C.3 Data policy

Catalogued third-party data is cited and, where licensing is unresolved, **never bundled**: the Fitvalu tables are referenced by locator and digest [Fit] (URL, file name, byte count and SHA-256 in the bibliography entry and in the repository source lock), and third-party PDFs are cited by locator without redistribution. Web catalogue records (Boyer’s pages [Boy], Brown’s note [Bro]) are used under dated locators recorded in the repository’s source locks. Bounded censuses are labelled with their frozen bounds and are never used as evidence for an unbounded statement; where a census executable has a recorded interface defect, only its frozen default output is cited.

### C.4 Manifest

The companion repository ships a manifest of SHA-256 digests covering every shipped source byte of the public package — scripts, certificates, transcripts, documentation and this manuscript’s sources; the compiled PDF (a deterministic function of the pinned sources on a fixed TeX distribution) and the regenerated results directory are deliberately unpinned. The original verifier has

a frozen expected output whose digest is pinned inside the verifier. A separate exact verifier covers the smooth-support catalogue and its own frozen expected receipt. The original verifier compares its output with the frozen expectation byte for byte. The smooth-support verifier compares exact JSON values, types and structure, independently of whitespace and object-key order. Both require their expected receipt and fail on its absence or on a failed comparison.

## C.5 AI assistance

The author used OpenAI Codex with substantial assistance in mathematical exploration, proposed arguments, code, exact computational checks, manuscript preparation and adversarial review. The author selected the research direction and is responsible for the submitted claims. Specific historical model versions are not reliably documented here and are not inferred from dates. AI-assisted reviews and computational replays are not independent specialist review or an end-to-end formal verification of the manuscript.

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